Help Me Do It on My Own: Sensory Data Analysis with jamovi and the SEDA Module

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Preface

This booklet was born from a simple conviction: we learn better to do things on our own when someone has taken the time to explain not only **how**, but also **why**.

The statistical analysis of sensory data has long been learned in R, through books, scripts, and tutorials that assumed a certain familiarity with programming, packages, and sometimes even a degree of resilience in the face of error messages. These tools have their strengths, their elegance, and remain indispensable. But they can also be a barrier for many practitioners who nevertheless work regularly with sensory data, without coming from the world of statistics or coding.

jamovi has partly changed the situation. By making methods accessible through its interface that were once reserved for more technical users, it has widened access to the statistical analysis of sensory data. But making a method clickable is not enough to make it understandable. A button without explanation remains a black box; a result without a reading framework remains fragile; a map without interpretation remains mute.

This is precisely the space this booklet seeks to occupy.

It replaces neither a full statistics course nor in-depth training in R. Its ambition is more focused: to give you, for each analysis, **enough understanding to work with discernment**. Enough to know what a confidence ellipse really represents. Enough to understand why a product may be absent from a descriptive table without thereby being "without a profile". Enough not to read a preference map as an absolute truth far from the products actually tested. Enough, above all, no longer to be entirely dependent on an interface that one uses without understanding.

The guiding thread of this book is therefore **autonomy**. Not the immediate autonomy of someone who already knows everything, but the gradual autonomy of someone who can move forward with reliable points of reference, connect the visible options in jamovi to their statistical logic, and interpret the outputs more accurately. In that sense, this booklet deliberately sits between two worlds: that of concrete practice in jamovi, and that of methodological understanding, which gives meaning to the results.

This booklet covers the **seven analyses** of the **SEDA** module, in the following order: **univariate characterisation of the product space**, **multivariate representation with confidence ellipses**, **CATA** data analysis, **JAR** data analysis, **Napping**, **free sorting**, and finally **preference mapping**. Together, they form a coherent progression: from fixed-list attribute methods to more holistic approaches, and then to hedonic analyses. Each method corresponds to a particular way of producing data, asking a question, and interpreting a result.

To maintain overall coherence, each chapter follows the same structure: an introduction that sets out the statistical problem, a section explaining what is really happening behind the interface, a detailed presentation of the options, a step-by-step procedure, and then keys for interpreting the tables and graphs. This choice is not merely editorial; it is pedagogical. It is intended to ensure that, from one chapter to the next, the reader finds stable reference points and can more easily transfer what has been learned.

This booklet is intended for sensory evaluation practitioners, engineers, students, teachers, researchers discovering jamovi, and more broadly for anyone who needs support between a course and a real field analysis.

Each chapter can be read independently. You can enter this booklet through the method that corresponds to the data you currently have. But the whole has also been conceived as a journey: the further one goes, the more one understands that sensory analysis is not a stack of separate techniques, but a family of complementary approaches, articulated through common logics of representation, description, and interpretation.

At heart, this book pursues a very simple idea: a good tool should not merely make it possible to obtain a result, but should help one understand what that result means. If these pages allow you to move from *"show me how to do it"* to *"now I can do it on my own"*, then they will have fully achieved their purpose.

1 Introduction

1.1 Presenting jamovi

1.1.1 What is jamovi?

In one sentence: *jamovi is a free, modern, and accessible statistics package, designed for both teaching and research.*

Developed by contributors from the JASP project, jamovi relies on R for its statistical calculations while providing a graphical interface that requires no code at all. In practical terms, its interface works like a dynamic spreadsheet[1]: the data are visible and editable in a sheet on the left, and the results update in real time on the right. This organisation makes the tool accessible to both beginners and experienced users.

One often underestimated advantage of jamovi is its ability to save everything in a single `.omv` file: the data, the analysis options selected, and the results. This greatly facilitates reproducibility and the sharing of work, two areas in which traditional spreadsheets show their limits.

A comparison with other common tools is useful for situating jamovi:

Feature	Excel / Google Sheets	jamovi
Data entry	✔	✔
Statistical analyses	✖(requires external tools)	✔(built in natively)
Reproducibility	⚠difficult to track	✔via the `.omv` format
Advanced modules	✖	✔via extensions
Open source	✖	✔

What about XLSTAT? XLSTAT is a paid-for Excel plug-in that adds advanced statistical functions. Although it is powerful, it remains proprietary, dependent on Microsoft Excel, and its cost may be a barrier for students or organisations with limited budgets. jamovi offers comparable features, and in some areas superior ones, while being entirely free and independent.

[1] A *spreadsheet* is a data-processing tool organised into rows and columns, such as Microsoft Excel or Google Sheets.

1.1.2 How do you install jamovi?

Installation is straightforward and requires no technical skills. jamovi is cross-platform: it works in the same way on Windows, macOS, and Linux. To install it, go to jamovi.org/download, choose your operating system, then run the downloaded file.

On the download page, two versions are offered. The **Solid** version is the most stable and thoroughly tested: it is the one to favour for teaching, practical classes, and research. The **Current** version includes the very latest features but may contain a few minor instabilities.

1.2 Modules in jamovi

In jamovi, a module is the equivalent of a package in R or Python: it adds specific features to the software, whether in the form of new types of analyses, graphs, or statistical models.

The jamovi module ecosystem is already rich and covers many needs:

Module	Main features
GAMLj	Linear, logistic, and hierarchical models
semlj	Structural equation models
jsq	Bayesian methods
PAMLj	Statistical power analysis
lsj	Interactive exercises from the *Learning Stats* manual
MEDA	PCA, CA, MCA, MFA (FactoMineR under jamovi)
SEDA	Sensory data analysis (SensoMineR under jamovi)

To install a module, simply open jamovi, go to the **Modules** tab in the top right-hand corner, select **jamovi library**, then choose the module you want and click **Install**. Some modules still under development may also be installed manually from a `.jmo` file.

1.3 The SEDA module

SEDA (*Sensory Evaluation Data Analysis*) is the jamovi module dedicated to the statistical analysis of sensory data. It is based on the R package *SensoMineR* and makes the main statistical methods used in sensory analysis available through a graphical interface — without requiring a single line of code.

Its menu is organised into three sections corresponding to three broad ways of producing sensory data.

The first section, **Fixed List of Attributes**, brings together methods in which subjects evaluate stimuli using a list of attributes defined in advance. This is the classic case of QDA (*Quantitative Descriptive Analysis*) data, where a trained panel scores the intensity of each attribute for each product. Here, SEDA offers two complementary analyses: one to **characterise the stimuli attribute by attribute** (by identifying which attributes differentiate them and which characterise each stimulus in particular), and another to **represent the sensory space in a multivariate way** (by producing a map of the stimuli enriched with confidence ellipses reflecting the stability of the positions according to panel composition). This section also includes the analysis of **CATA** (*Check-All-That-Apply*) data, in which respondents tick from a list the terms they feel are present in a stimulus, and the analysis of **JAR** (*Just About Right*) data, in which respondents indicate whether each attribute is too strong, too weak, or just about right — making it possible to estimate the *penalties* these deviations impose on liking.

The second section brings together **holistic approaches**, in which subjects evaluate the stimuli globally, without any predefined list of attributes. Two methods are covered: **Napping** (*projective mapping*), in which each subject positions the stimuli on a rectangle according to their perceived similarities, and **free sorting** (*sorting task*), in which each subject freely groups the stimuli into categories. In both cases, SEDA produces a consensual representation of the stimuli by combining the individual maps — by Multiple Factor Analysis for Napping, and by Multiple Correspondence Analysis for free sorting — and may complement this representation with an automatic classification of the stimuli.

The third section, **Hedonic Data**, is dedicated to **liking data**. It contains **preference mapping**, which combines an already constructed representation space (for example, from a PCA) with individual liking scores in order to produce a response surface indicating, for each area of the space, the proportion of consumers who would theoretically respond favourably to it. This analysis is the natural extension of the previous analyses: once the sensory space has been constructed and interpreted, the preference map makes it possible to locate the product optimum within that space.

This booklet covers all seven of these analyses, in the order of the SEDA menu. Throughout the rest of this booklet, the terms *product* and *stimulus* mean the same thing and are used interchangeably.

1.4 Some useful references

- Official website: jamovi.org
- Online documentation
- jamovi user manual
- Navarro, D., Foxcroft, D., & Meunier, J.-M. (2020). *Apprentissage des statistiques avec Jamovi*.

- Love, J., Selker, R., Marsman, M. et al. (2019). *JASP: Graphical statistical software for common statistical designs*. Journal of Statistical Software, 88, 1–17.
- Navarro, D., & Foxcroft, D. (2018). *Learning statistics with jamovi*.

- Community and modules: forum.jamovi.org

2 QDA Data — Characterization of the Sensory Space

2.1 Introduction

This analysis is designed for **QDA** (*Quantitative Descriptive Analysis*) data, when several subjects evaluate several stimuli using a fixed list of sensory attributes. It answers two complementary questions: **which attributes structure the sensory space as a whole** and **which attributes characterize each stimulus in particular**. More specifically, the function returns two families of results: a ranked list of the attributes that differentiate the set of stimuli, then, for each stimulus, a list of the attributes that characterize it.

Why not use a standard alternative?

A simple comparison of means by stimulus can be misleading if subjects do not all use the rating scale in the same way. Here, the analysis takes into account both the product effect (**Stimulus Effect**) and the subject effect (**Subject Effect**), which makes it possible to identify discriminating attributes and profiles more *robustly* than with a purely descriptive reading.

To test this function, simply load the *sensochoc* data available with the SEDA module — they appear in the list of example datasets when opening a new file in jamovi (*see* section 7).

2.2 Understanding what happens under the hood

2.2.1 Data checking and preparation

The analysis first checks that the variable placed in the **Stimulus Effect** field and the variable placed in the **Subject Effect** field can be treated as factors, and that all variables placed in **Sensory Attributes** are numeric. If a sensory attribute is not numeric, the analysis stops with an error message.

2.2.2 An attribute-by-attribute analysis

A simple way to understand the computation is to say that the function proceeds **attribute by attribute** using an analysis of variance (ANOVA). For each variable in **Sensory Attributes**, it first evaluates whether the *product* effect is significant, while taking the *subject* effect into account. This first step acts as a global filter: it determines which attributes truly structure the sensory space.

2.2.3 Selection of discriminating attributes

Attributes whose significance is below the threshold defined by **Significance threshold (%)** are retained in the first results table. The default threshold is **5%**, which makes it a good starting point for a cautious reading. If no attribute passes this threshold, the function stops and indicates that no sensory attribute is discriminating at the selected threshold.

2.2.4 Stimulus-by-stimulus description

Once the globally discriminating attributes have been identified, the function examines, for each stimulus, which attributes significantly characterize it. Each stimulus is compared with a global reference, which makes it possible to identify attributes positively or negatively associated with that stimulus. The global reference corresponds to the mean of the stimuli on each attribute, that is, the average stimulus. The result takes the form of a second table grouped by stimulus.

2.2.5 The role of adjusted means

The **Adjusted mean** column can be understood as an estimate of the stimulus mean on an attribute, corrected for the subject effect. This column makes it possible to read the perceived level of a stimulus without being limited to a raw mean that may be influenced by subjects who are more "generous" or more "severe" in their ratings.

> When the tasting design is complete, that is, when each subject evaluates all stimuli, the adjusted mean is equal to the arithmetic mean. It differs from it when data are missing; in that case, the adjusted mean is more reliable.

2.2.6 What it means when a stimulus is absent from the second table, Sensory Description of the Stimuli

A stimulus may not appear in the second table if no attribute significantly characterizes it at the chosen threshold. This does not mean that it has “no profile”, but rather that it does not differ clearly enough from the mean of the full set of stimuli on the tested attributes.

2.3 Expected data structure

The data must contain:

- one column identifying the **stimuli**;
- one column identifying the **subjects**;
- several numeric columns corresponding to the **sensory attributes**.

Each row corresponds to the evaluation of one stimulus by one subject. If subjects evaluate the stimuli over several repetitions or several sessions, each evaluation must appear on a separate row.

Before running the analysis in jamovi, check in the **Data** tab that:

- the variable intended for **Stimulus Effect** is categorical;
- the variable intended for **Subject Effect** is categorical;
- all variables intended for **Sensory Attributes** are numeric.

2.4 Interface guide

The analysis appears in the menu **SEDA > Fixed List of Attributes > Characterization of the Stimulus Space**. The interface includes three variable drop zones, a **Read me before running** checkbox, and a numeric field, **Significance threshold (%)**.

2.4.1 Variables to complete

2.4.1.1 Stimulus Effect

Role: identifies the stimuli/products to compare. **Impact on the computation:** this variable defines the groups studied in the global analysis and in the stimulus-by-stimulus description. **Practical tip:** use explicit labels, because they directly structure the reading of the second table.

2.4.1.2 Subject Effect

Role: identifies the subjects. **Impact on the computation:** this variable is used to take systematic differences between subjects into account when estimating stimulus effects. **Practical tip:** check that the same subject keeps exactly the same identifier throughout the file.

2.4.1.3 Sensory Attributes

Role: groups together the sensory attributes to be analyzed. **Impact on the computation:** the function applies the same analysis logic to each of these attributes; only numeric attributes are accepted. **Practical tip:** include all relevant attributes from your QDA; the statistical selection will then be performed through **Significance threshold (%)**.

2.4.2 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

2.4.3 Significance threshold: Significance threshold (%)

Role: sets the *p-value* threshold used to select discriminating attributes and the attributes characteristic of each stimulus. **Impact on the computation:** this threshold controls both which attributes appear in **Identification of the Sensory Dimensions** (global test) and which attributes appear in **Sensory Description of the Stimuli** (stimulus-level tests). A low threshold produces more conservative results; a higher threshold retains more attributes in both tables. **Practical tip:** the default value is **5%**, a good starting point for a first trial. If the results are too sparse, try **10%**, then possibly **20%** for exploratory work.

2.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical, that the subject variable is categorical, and that the sensory attributes are numeric.
3. Open **SEDA > Fixed List of Attributes > Characterization of the Stimulus Space**.
4. Place the stimulus variable in **Stimulus Effect**.
5. Place the subject variable in **Subject Effect**.
6. Place all sensory attributes in **Sensory Attributes**.
7. Leave **Significance threshold (%)** at **5%** for a first trial.

8. First read **Identification of the Sensory Dimensions** to identify the attributes that structure the sensory space.
9. Then read **Sensory Description of the Stimuli** to understand the characteristics specific to each stimulus.
10. If no results appear, or if the second table is very sparse, test a slightly broader threshold.

2.6 Interpretation of results

2.6.1 Identification of the Sensory Dimensions

This first table lists the sensory attributes that **significantly discriminate** the stimuli, sorted from the most discriminating to the least discriminating. It is the entry point for interpretation: it indicates which attributes structure the sensory space as a whole.

Element	Meaning
(first column)	Discriminating sensory attribute
Vtest	Standardized measure of the discriminating effect — the higher the value, the stronger the signal
p	P-value of the global test — the lower it is, the more clearly the attribute differentiates the stimuli

The first rows of the table correspond to the most structuring attributes.

> **Point of vigilance:** an attribute absent from this table is not sufficiently discriminating at the chosen threshold. If no attribute is retained, the function displays an error message and does not produce interpretable results.
>
> **Point of vigilance:** if only one variable is placed in **Sensory Attributes**, which will **never** happen in practice, it will be renamed with the number **1** in the table. The logic of interpretation remains the same: first read the global discrimination, then the stimulus-by-stimulus characterization.

2.6.2 Sensory Description of the Stimuli

This second table describes each stimulus through the attributes that significantly characterize it. Rows are grouped by stimulus, and the most salient attributes appear first within each group.

Element	Meaning
(first column)	Name of the stimulus
Descriptor	Characteristic sensory attribute
Coefficient	Estimated deviation from the global reference — positive if the stimulus is perceived more intensely, negative otherwise
Adjusted mean	Adjusted mean of the stimulus on this attribute, corrected for the subject effect
p	P-value of the characterization test
Vtest	Standardized intensity of the association — positive = over-represented, negative = under-represented

- A positive **V-test** suggests that the stimulus is perceived as more intense than the reference, that is, the mean, on this attribute.
- A negative **V-test** suggests the opposite — and is just as informative.
- Use **Adjusted mean** to read the sensory level, but never interpret it without **p** and **Vtest**.

  Point of vigilance: a stimulus absent from the table does not have "no profile"; it is simply not significantly different from the mean of the full set of stimuli at the chosen threshold.

In the current configuration of the function, no graphical output is produced. Interpretation is based entirely on the two result tables.

2.7 Example: the sensochoc dataset

The **sensochoc** dataset describes 6 chocolates evaluated by 29 subjects on 14 sensory attributes. It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

2.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimulus Effect	Product
Subject Effect	Panelist

Option	Value
Sensory Attributes	all numeric sensory attributes
Significance threshold (%)	5

2.7.2 Analysis of the *Identification of the Sensory Dimensions* table

With these settings, all 14 attributes emerge as discriminating at the 5% threshold. The table is sorted by decreasing V-test: the attributes at the top are those that most clearly differentiate the chocolates from one another. *MilkF* comes first (V-test = 16.40), followed by *CocoaF* (13.48) and *Bitterness* (13.34). At the bottom of the table, *Sticky* (V-test = 3.56) and *Granular* (4.37) are significant but less structuring. This ranking provides a first reading of the sensory space: the contrasts between chocolates are based above all on milky, cocoa, and bitter dimensions.

2.7.3 Analysis of the *Sensory Description of the Stimuli* table

The second table makes it possible to read the profile of each chocolate relative to the overall mean of all six chocolates.

choc1 is characterized by a highly contrasted profile. For this chocolate, on the positive side, *Bitterness* has the highest V-test (9.95, coefficient +2.46) and an adjusted mean of 7.07, clearly above the overall mean. *CocoaF* (V-test = 8.48), *Astringency* (7.08), and *Acidity* (6.14) complete this profile. On the negative side, *Sweetness* (V-test = -8.29, coefficient -1.95, adjusted mean 3.14) and *MilkF* (V-test = -8.70) indicate a chocolate that is very low in sweetness and very low in milky notes.

choc3 presents the opposite profile. It is strongly characterized by *MilkF* (V-test = 16.52, the highest in the table), *Caramel* (11.52), *Sweetness* (10.39), and *Melting* (9.19). Conversely, it is negatively characterized by *Bitterness* (-12.42), *Crunchy* (-12.49), and *CocoaF* (-13.33). It is the milkiest, sweetest, and most melting chocolate, and the least bitter and cocoa-like. Given the number of significant sensory attributes, **choc3** is certainly the chocolate whose profile differs most clearly from the mean.

choc2 stands out mainly through *Crunchy* (V-test = 6.92) and a lower perception of *MilkF* (-5.16). **choc6** is also *Crunchy* (5.35) and slightly more *Sweetness* (2.41), but less *Melting* (-3.08).

choc4 shows a moderate and weakly contrasted profile — except for *MilkF*, no V-test exceeds 4 in absolute value — which suggests a chocolate close to the general sensory mean on most attributes.

choc5 has very few salient characteristics: only *CocoaA*, *Crunchy*, and *CocoaF* appear slightly on the positive side, and *Sticky* on the negative side. Its profile is the least distinctive of the six chocolates at the 5% threshold.

2.7.4 Synthetic interpretation

This example shows that the six chocolates cover a broad sensory space, with a very bitter and intense chocolate at one extreme (**choc1**) and a milky, sweet, and melting chocolate at the other (**choc3**). Between these two poles, the other chocolates occupy more intermediate positions. The first table identifies the attributes that globally structure the sensory space, while the second specifies the profile specific to each stimulus. The joint reading of these two outputs is the main contribution of the function.

> **What we take away from this example:** the analysis combines a global reading of discriminating attributes and a local reading of stimulus profiles, making it possible to characterize the sensory space in detail.

2.8 Practical tips

- Start with **Significance threshold (%) = 5%**.
- If no attribute is retained, test **10%**, then **20%** for exploratory work.
- Always read the global table before the stimulus-by-stimulus table.
- Do not confuse a globally discriminating attribute with an attribute that strongly characterizes **all** stimuli: an attribute can differentiate stimuli from one another without all of them being strongly associated with it.
- A negative **V-test** is informative: it indicates that a stimulus is perceived as less intense than the mean on an attribute, which can be a sensory characteristic in its own right.
- Use clear and consistent stimulus names to make the results easier to read.
- If your panel includes repetitions, keep all rows in the file.
- Do not overinterpret the absence of a stimulus from the second table: it may simply be located close to the general sensory mean.

2.9 In summary

This analysis first identifies the sensory attributes that globally discriminate the stimuli, then describes each stimulus through the attributes that distinguish it from the global reference, that is, the mean. It takes into account both the stimulus effect and the subject effect, which makes it more informative than a simple comparison of raw means. Its main value is that it combines a **global** reading of the sensory space and a **local** stimulus-by-stimulus reading, in two directly interpretable tables.

3 QDA Data — Multivariate Representation with Ellipses

3.1 Introduction

This analysis is designed for **Quantitative Descriptive Analysis (QDA)** data, that is, when several subjects evaluate several stimuli using a fixed list of sensory attributes. It extends the univariate characterization of the sensory space by providing a multivariate representation of the stimuli and their variability. It answers two complementary questions: **which stimuli are sensorially similar** and **is this proximity stable despite variability between subjects**? Its central output is a map of the stimuli enriched with **ellipses**, which visualize not only the position of the stimuli, but also the robustness of this position under panel resampling.

Why not use a standard alternative?

A PCA applied directly to the raw data may mix two sources of variation: the **stimulus** effect and the **subject** effect. In sensory analysis, some subjects systematically rate higher or lower than others, or use the scale more or less broadly. This analysis takes this subject/stimulus structure into account to produce a representation better suited to QDA data.

3.2 Understanding what happens under the hood

3.2.1 Selection of sensory attributes

Before constructing the map, the function identifies the attributes that discriminate the stimuli sufficiently. This selection depends on the value of **p-value of the Stimulus effect (%)**. The stricter the threshold, the more streamlined the map will be; the broader it is, the more exhaustive the representation will be, although sometimes more crowded. The default value is 20%, which makes it a good starting point for exploration.

3.2.2 Correction of the subject effect

A simple way to understand the next step is to say that the analysis reconstructs **mean sensory profiles by stimulus**, while taking differences between subjects into account. Two options are involved here:

- **Center by subject** corrects differences in average level between subjects;
- **Scale by subject** corrects differences in the amplitude with which subjects use the scale.

The objective is to prevent a very generous, very severe, or very "spread-out" subject from artificially distorting the map of the stimuli.

3.2.3 Principal component analysis

A **PCA** is then performed on the resulting table of sensory profiles. It produces the stimulus map, the sensory attribute map, the eigenvalue table, and the automatic description of the dimensions. This factor space is then used as the reference space for projecting the virtual panels.

3.2.4 Virtual panels and supplementary individuals

The key point of the function is the use of **resampling**. In PCA, it is possible to distinguish between **active individuals** — those used to construct the axes — and **supplementary individuals** — those projected afterwards, without modifying the axes. This is precisely the mechanism that makes it possible to construct the ellipses.

At each simulation, a subset of subjects is drawn with replacement from the original panel, new mean stimulus profiles are recalculated, and then projected into the factor space as supplementary individuals. The dispersion of these projections around each stimulus is then summarized by an ellipse.

These ellipses therefore describe the variability of the **mean position of the stimuli when the composition of the panel changes**, not the simple raw dispersion of individual ratings.

> **Note:** the computation may take a few seconds, because it relies on a large number of virtual panels. During this computation, it is better to avoid changing other options so as not to restart the procedure unnecessarily.

3.2.5 Comparison between stimuli

The function also computes a comparison between pairs of stimuli, presented as a matrix of **p-values**. This output complements the visual reading of the ellipses by adding a numerical measure of the difference between stimuli.

3.2.6 Automatic description of the dimensions

Finally, the function provides an automatic description of the axes, filtered according to the value of **Significance threshold (%)**. It helps identify which attributes are most strongly associated with each dimension and gives a *sensory* meaning to the axes.

3.3 Expected data structure

The data must contain:

- one column identifying the **stimuli**;
- one column identifying the **subjects**;
- several numeric columns corresponding to the **sensory attributes**.

Each row corresponds to the evaluation of one stimulus by one subject. If a subject evaluates the same stimulus several times (repetitions or sessions), each evaluation must appear on a separate row.

Before running the analysis in jamovi, check in the **Data** tab that:

- the variable intended for **Stimulus Effect** is categorical;
- the variable intended for **Subject Effect** is categorical;
- all variables intended for **Sensory Attributes** are numeric.

3.4 Interface guide

The analysis appears in the menu **SEDA > Fixed List of Attributes > Representation of the Stimulus Space**. The interface includes three variable drop zones, integrated help, and then three collapsible menus: **Graphic Options**, **Resampling Options**, and **Numerical Indicators**.

3.4.1 Variables to complete

3.4.1.1 Stimulus Effect

Role: identifies the stimuli to be represented on the map. **Impact on the computation:** each level of this variable becomes a mean point and, where applicable, an ellipse on the map. **Practical tip:** use clear labels, because they will appear directly in the graphics.

3.4.1.2 Subject Effect

Role: identifies the subjects. **Impact on the computation:** this variable is essential for correcting the subject effect and for generating the virtual panels used in the bootstrap. **Practical tip:** check that the same subject keeps exactly the same identifier throughout the file.

3.4.1.3 Sensory Attributes

Role: groups together the sensory attributes used to construct the factor space. **Impact on the computation:** only numeric attributes are accepted; the number of calculable dimensions depends on the number of retained attributes. **Practical tip:** start by including all relevant sensory attributes, then refine the selection with **p-value of the Stimulus effect (%)**.

3.4.2 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

3.4.3 Standardization of the Sensory Attributes: Scale to unit variance

Role: standardizes the sensory attributes when performing the PCA. **Impact on the computation:** when checked, each attribute contributes equally to the construction of the axes; when unchecked, the most dispersed attributes have a greater influence on the construction of the axes. **Practical tip:** leave this option checked in most analyses. It is **enabled by default**.

3.4.4 Selection of the Sensory Attributes: p-value of the Stimulus effect (%)

Role: sets the threshold for selecting attributes according to their relationship with the stimulus effect. **Impact on the computation:** a low threshold retains few attributes and produces a more readable map; a higher threshold retains more attributes and gives a broader view. **Practical tip:** the default value is 20%, a good starting point for exploration. For a stricter version, start at 5%.

3.4.5 Graphic Options

3.4.5.1 Components to Plot — X-axis and Y-axis

Role: choose the dimensions displayed on the horizontal and vertical axes. **Impact on the computation:** they do not change the PCA itself, but they modify the displayed factor plane. The selected dimensions must exist in the analysis, otherwise the function generates an error. **Practical tip:** start with X-axis = 1 and Y-axis = 2, the default values.

3.4.5.2 Individual variability around stimuli

Role: adds a graph showing the dispersion of individual evaluations around each stimulus. **Impact on the computation:** it does not add a new analysis, but provides an additional graphical output useful for assessing consensus between subjects. **Practical tip:** activate this option if you want to distinguish a consensual stimulus from a stimulus perceived more variably. The default value is **disabled**.

3.4.5.3 Variability around sensory attributes

Role: adds a graph showing the variability of sensory attributes under resampling. **Impact on the computation:** it provides a graphical output useful for assessing the stability of attribute interpretation. **Practical tip:** activate this option when you want to comment on the robustness of the axes. The default value is **disabled**.

3.4.6 Resampling Options

3.4.6.1 Number of subjects

Role: defines the number of subjects drawn in each virtual panel. **Impact on the computation:** the closer this value is to the actual panel size, the more similar the simulated panels will be to it. **Practical tip:** use the actual panel size as a starting point. The default value is 20.

3.4.6.2 Number of panels

Role: sets the total number of simulated virtual panels. **Impact on the computation:** the larger this number, the more stable the ellipses will be from one run to another, but computation time increases. **Practical tip:** 300 is a good compromise to start with; for a final rendering, it may be useful to increase this number to 500. The default value is 300.

3.4.6.3 Threshold (%)

Role: defines the level used to draw the ellipses. **Impact on the computation:** a low threshold produces wider ellipses corresponding to a higher confidence level. **Practical tip:** leave 5% for a first trial. This is the default value.

3.4.6.4 Center by subject

Role: centers responses subject by subject. **Impact on the computation:** corrects differences in average level between subjects and often improves the comparison between profiles. **Practical tip:** leave this option checked in almost all analyses. It is **enabled by default**.

3.4.6.5 Scale by subject

Role: standardizes responses subject by subject. **Impact on the computation:** corrects differences in amplitude between subjects, but modifies the initial structure more strongly than centering alone. **Practical tip:** leave this option unchecked at first. It is **disabled by default**.

3.4.7 Numerical Indicators

3.4.7.1 Significance threshold (%)

Role: defines the significance threshold for the automatic description of the dimensions. **Impact on the computation:** the stricter the threshold, the more selective the automatic description will be. **Practical tip:** the default value, 5%, is well suited to a first reading.

3.4.7.2 Number of dimensions

Role: defines the number of dimensions to calculate and describe. **Impact on the computation:** this value is limited by the dimensions that can actually be calculated by the PCA. **Practical tip:** the default value, 2, is appropriate for an initial reading.

3.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical, the subject variable is categorical, and the sensory attributes are numeric.
3. Open **SEDA > Fixed List of Attributes > Representation of the Stimulus Space**.
4. Place the stimulus variable in **Stimulus Effect**.
5. Place the subject variable in **Subject Effect**.
6. Place all sensory attributes in **Sensory Attributes**.
7. Leave **Scale to unit variance** checked.
8. Set **p-value of the Stimulus effect (%)** to 20% for an initial exploration, or to 5% for a stricter selection.
9. Keep X-axis = 1 and Y-axis = 2.
10. Leave **Center by subject** checked and **Scale by subject** unchecked at first.
11. Set **Number of subjects** to the panel size or a close value, leave **Number of panels** at 300 and **Threshold (%)** at 5.
12. Activate **Individual variability around stimuli** and/or **Variability around sensory attributes** according to your needs.
13. In **Numerical Indicators**, keep the default values for a first trial.
14. First read **Representation of the Stimuli with Ellipses**, then **Eigenvalue Decomposition**, then **Automatic Description of the Dimensions**.
15. Then return to the settings to test the stability of your conclusions.

3.6 Interpretation of results

3.6.1 Representation of the Stimuli with Ellipses

This is the central graph of the function. Each stimulus is represented by a mean point in the factor space, surrounded by an ellipse. The axes come from the PCA, and the percentage displayed on each axis indicates the share of variance explained.

Element	Meaning
Two close stimuli	Similar sensory profiles
Strongly overlapping ellipses	Weakly significant difference (with respect to a change in panel)
Small ellipse	Position stable across simulated panels
Large ellipse	More uncertain position
Clearly separated ellipses	More significant difference

Point of vigilance: the ellipses describe the variability of the mean position of the stimuli under panel resampling. They do not directly represent the raw dispersion of individual ratings.

3.6.2 Representation of the Stimuli

This map shows the relative positions of the stimuli without emphasizing the ellipses. It is useful for quickly identifying the main groupings and oppositions between stimuli.

3.6.3 Representation of the Sensory Attributes

This map shows the sensory attributes in the factor space. It is read using three simple cues:

- **direction**: toward which stimuli the attribute "pulls" the structure;
- **length**: the longer an attribute is, the better it is represented on the plane;
- **the angle between two attributes**: small angle = positive correlation; angle close to 180° = opposition; angle close to 90° = near-independence.

This is the map that gives sensory meaning to the axes.

3.6.4 Individual variability around stimuli *(available only if the option is activated)*

This output shows the dispersion of individual evaluations around the mean point of each stimulus. A compact cloud indicates strong consensus; a dispersed cloud indicates a more heterogeneous perception between subjects.

3.6.5 Variability around sensory attributes *(available only if the option is activated)*

This output shows the dispersion of the position of sensory attributes in the virtual panels. A compact cloud means that the attribute is stable; a dispersed cloud indicates that its interpretation is more fragile and depends more strongly on the composition of the panel.

3.6.6 Eigenvalue Decomposition

This table gives, for each dimension, the eigenvalue, the percentage of variance explained, and the cumulative percentage. If the first two dimensions explain a large share of the variance, the 2D map summarizes the data well. Otherwise, it is prudent to explore other planes, for example Dim 1 / Dim 3.

3.6.7 Automatic Description of the Dimensions

This textual output identifies the attributes most strongly linked to each dimension, filtered by **Significance threshold (%)**. It helps name the axes in sensory terms and complements the attribute map well.

3.6.8 Product-by-product comparison table *(available only if Number of subjects > 1)*

This p-value matrix compares each pair of stimuli in the factor space. A low p-value suggests a clearer difference between two stimuli; a high p-value suggests a more plausible proximity. This table complements the visual reading of the ellipses, especially when their overlap is difficult to interpret by eye.

3.7 Example: the sensochoc dataset

The **sensochoc** dataset describes 6 chocolates evaluated by 29 subjects on 14 sensory attributes. It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

3.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimulus Effect	Product
Subject Effect	Panelist
Sensory Attributes	all sensory attributes
Scale to unit variance	Checked
p-value of the Stimulus effect (%)	20
X-axis / Y-axis	1 / 2
Number of subjects	20
Number of panels	300
Threshold (%)	5
Center by subject	Checked
Scale by subject	Unchecked
Significance threshold (%)	5
Number of dimensions	2

Note: The results presented in this section are obtained from the default parameter values. The number of subjects making up the panel should be set to 29, the actual number of subjects in the experiment.

3.7.2 Analysis of the *Representation of the Stimuli with Ellipses* graph

The map shows the 6 chocolates in the Dim. 1 / Dim. 2 plane. The first dimension alone accounts for 88.79% of the total variance — this 2D plane is therefore a very faithful representation of the data. The second dimension explains an additional 7.58%.

On the horizontal axis (Dim. 1), **choc1** and **choc3** occupy the two opposite extremes: **choc1** on the left (bitter, cocoa side), **choc3** on the right (milky, sweet side). These are the two chocolates whose profiles are most clearly opposed, which is consistent with the results of the previous chapter. **choc2**, **choc5**, and **choc6** are grouped in the center, close to the origin. **choc4** is located slightly above the horizontal axis, between choc1 and the central group.

The ellipses provide information about the robustness of these positions. The ellipses of **choc2**, **choc5**, and **choc6** overlap widely. In other words, if the composition of the panel were changed, their order could be reversed. Conversely, **choc1** and **choc3** have ellipses clearly separated from all the others: their position is stable, and their difference from the other chocolates can be considered reliable.

3.7.3 Analysis of the *Hotelling Test* table

The *p-value* table quantitatively confirms what the ellipses show visually. The pairs **choc2 / choc5** ($p = 0.76$) and **choc5 / choc6** ($p = 0.16$) have high p-values: one cannot conclude that there is a significant difference between these chocolates in the sensory space. The pair **choc2 / choc6** is borderline ($p = 0.04$). By contrast, all comparisons involving **choc1** or **choc3** give p-values of 0.00: these two chocolates differ significantly from all the others.

3.7.4 Analysis of the *Representation of the Sensory Attributes* graph

The 14 attributes are distributed very clearly around the horizontal axis. On the right are grouped *MilkF*, *Sweetness*, *Caramel*, *Vanilla*, *MilkA*, and *Melting* — this is the profile of **choc3.** On the left are grouped *CocoaF*, *CocoaA*, *Bitterness*, *Astringency*, *Acidity*, *Crunchy*, and *Granular* — this is the profile of **choc1**. The horizontal axis therefore clearly corresponds to an opposition between **sweet milk chocolate / bitter dark chocolate**.

The length of the arrows also indicates the quality of representation of each attribute on this plane. *Crunchy* has a shorter arrow pointing downward, reflecting part of its variability on the second dimension — confirming that this attribute is less stable in subjects' perception than *CocoaF* or *MilkF*, whose arrows almost reach the edge of the circle.

3.7.5 Analysis of the *Automatic Description of the Dimensions* table

The automatic description of the dimensions confirms these readings. On Dim. 1, the most positively correlated attributes are *MilkF* (correlation 0.86), *Sweetness* (0.66), and *Caramel* (0.65); the most negatively correlated are *CocoaF* (-0.79) and *Bitterness* (-0.70). On Dim. 2, *Sticky* (0.59), *Acidity* (0.58), and *Astringency* (0.57) appear on the positive side, while *Melting* (-0.15) is slightly negative.

3.7.6 Synthetic interpretation

The sensory space of the chocolates is very largely structured by a single strong opposition, carried by the first dimension: on one side, a milky, sweet, and melting profile; on the other, a bitter, cocoa, and more textured profile. The stimulus map and the attribute map converge toward the same overall reading. The ellipses additionally show that not all stimuli are differentiated with the same robustness. Finally, the Hotelling test confirms that some proximities observed on the map do not correspond to clearly established differences.

> **What we take away from this example:** the analysis does not only show where the stimuli are located in the sensory space, but also how robustly they can be distinguished.

3.8 Practical tips

- Always start with X-axis = 1 and Y-axis = 2.
- Leave **Scale to unit variance** checked in most cases.
- Use **Center by subject** almost systematically.
- Activate **Scale by subject** only if you have a good reason to think that subjects use the scale with very different amplitudes.
- Start with 300 simulated panels; increase to 500 or more for a more stable final version.
- Set **p-value of the Stimulus effect (%)** between 5% and 20% depending on whether you want a streamlined or more exhaustive map.
- Never interpret the stimulus map without looking at the attribute map.
- Do not conclude too quickly that there is a difference when the ellipses overlap strongly.
- Consult **Product-by-product comparison table** to complement the visual reading when the overlap is ambiguous.
- If the first two dimensions summarize little variance, explore other factor planes.

3.9 In summary

This analysis produces a multivariate representation of stimuli suited to QDA data, while taking the subject effect into account in the construction of sensory profiles. It enriches the classical factor map with ellipses obtained by resampling, making it possible to assess the robustness of the observed positions. It therefore shows not only **where the stimuli are located**, but also **how confidently they can be located there**.

4 CATA Data — Analysis of Checked Terms

4.1 Introduction

This analysis is designed for **CATA** (*Check-All-That-Apply*) data, that is, when subjects check, from a predefined list, all the terms they consider appropriate to describe a stimulus. It answers two complementary questions: **which terms characterize each stimulus** and **how the stimuli are globally organized according to their CATA profiles**. The function produces a contingency table, a description of the stimuli according to the CATA terms, as well as a **correspondence analysis** followed by an **automatic classification** of the stimuli.

Why not use a standard alternative?

CATA data are originally binary responses by subject and by term. A classical comparison of means is therefore not the most suitable tool for representing the global structure of these data. The function first aggregates the responses into a **stimuli $\times$ terms contingency table**, then applies methods suited to this structure: frequency description, correspondence analysis, and, when possible, classification.

4.2 Understanding what happens under the hood

4.2.1 Construction of the contingency table

The first step consists of grouping the responses by stimulus. The function sums the columns placed in the **CATA** field for each level of **Stimulus Effect**. This produces a table where rows are stimuli, columns are CATA terms, and each cell indicates how many times a term was checked for a given stimulus. This table is the basis for all subsequent outputs.

> **Note:** this table can be copied into a spreadsheet file.

4.2.2 Description of stimuli by terms

From this contingency table, the function applies a frequency description procedure based on the threshold value defined in **Significance threshold (%)**. For each stimulus, it identifies the terms that are significantly more or less frequent than in the whole table. This logic is close to a statistical description of profiles, but adapted here to *checking* frequencies rather than to continuous ratings. The same threshold is then applied to the description of clusters.

> A term absent from the description table is not significantly associated with a stimulus at the chosen threshold. This may mean that it is checked fairly evenly across stimuli.

4.2.3 Correspondence analysis

If at least **three** variables are placed in the **CATA** field, the function performs a **correspondence analysis** on the contingency table. This method simultaneously represents the stimuli and the terms in the same factor space. Two stimuli that are close on the map tend to share similar CATA profiles; two terms that are close tend to be checked for similar stimuli. The axes summarize the main profile oppositions between stimuli and terms.

4.2.4 Automatic classification of stimuli

Once the correspondence analysis has been computed, the function then runs a **hierarchical clustering on principal components** with an automatically determined number of groups. It produces a map of the stimuli according to their clusters, then a description of the clusters using the same significance threshold as for the description of stimuli.

4.2.5 What happens if the number of terms is insufficient

The function requires at least **two** variables in the **CATA** field to start the analysis. Below that, it stops with a message indicating that the number of factors is too small. With exactly **two** terms, the tables can be produced, but the representation graphs and classification are not computed: they require at least **three** CATA variables.

4.3 Expected data structure

The data must contain:

- one categorical column identifying the **stimuli**;

- several numeric columns corresponding to the **CATA terms**. In practice, these columns are most often coded as **0/1**, since the function sums them by stimulus to construct the contingency table.

Each row corresponds to the evaluation of one stimulus by one subject. If 40 subjects evaluate 6 stimuli, the file generally contains one row per subject and per stimulus, with one column per *CATA* term.

Before running the analysis in jamovi, check in the **Data** tab that:

- the variable intended for **Stimulus Effect** is categorical;
- the variables intended for **CATA** are numeric and contain coherent values, most often 0s and 1s;
- you have selected at least **two** variables (items) in the **CATA** field, and ideally **three or more** if you want to obtain the graphs and the classification.

4.4 Interface guide

The analysis appears in the menu **SEDA > Fixed List of Attributes > Analysis of CATA Data**. The interface includes two variable drop zones, a **Read me before running** checkbox, and a numeric field, **Significance threshold (%)**.

4.4.1 Variables to complete

4.4.1.1 Stimulus Effect

Role: identifies the stimuli to compare. **Impact on the computation:** this variable defines the rows of the contingency table and organizes all descriptive and graphical outputs. **Practical tip:** use clear labels, because they will appear in the tables and in the graphs.

4.4.1.2 CATA

Role: groups together the terms from the CATA list. **Impact on the computation:** these variables define the columns of the contingency table; they are then used for the description of stimuli, correspondence analysis, and, when possible, classification. Only numeric variables are accepted. **Practical tip:** select all relevant CATA columns. At least **two** variables are required to run the analysis, and at least **three** are required to obtain the graphs and the classification.

4.4.2 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

4.4.3 Significance threshold: Significance threshold (%)

Role: defines the p-value threshold used to select the terms characteristic of stimuli and clusters. **Impact on the computation:** a low threshold produces more conservative results; a higher threshold retains more terms in the descriptive tables. The same threshold applies to the description of individual stimuli and to the description of clusters. **Practical tip:** the default value is 5%, a good starting point for a first trial. If the tables are too sparse, try 10%, then possibly 20% for exploratory work.

4.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical and that the CATA variables are numeric.
3. Open **SEDA > Fixed List of Attributes > Analysis of CATA Data**.
4. Place the stimulus variable in **Stimulus Effect**.
5. Place the CATA terms in the **CATA** field.
6. Check that you have selected at least **two** terms, and preferably **three or more** if you want the graphs.
7. Leave **Significance threshold (%)** at 5% for a first trial.
8. Start by reading the contingency table to check that the frequencies make sense.
9. Then read the description of stimuli to identify the most characteristic terms.
10. If at least three terms have been provided, consult the **Representation of the Stimuli and the CATA** map, then **Representation of the Stimuli According to Clusters**, and finally the description of clusters.
11. If the description tables are too empty, slightly increase **Significance threshold (%)**.

4.6 Interpretation of results

4.6.1 Contingency table

This first table crosses stimuli in rows and CATA terms in columns. Each cell indicates the number of times a term was checked for a given stimulus.

Element	Meaning
High value in a cell	The term was often checked for this stimulus
Row with several high values	The stimulus is associated with several frequent terms
Column high for almost all stimuli	Frequent term, but not necessarily discriminating

Point of vigilance: a very frequent term is not automatically a characteristic term. If it is checked for almost all stimuli, it will contribute little to distinguishing them.

4.6.2 Description of the Stimuli According to CATA

This table describes each stimulus through the terms that are significantly associated with it, grouped by stimulus. Terms are sorted by decreasing V-test within each group.

Element	Meaning
(first column)	Stimulus concerned
CATA	Characteristic term
Intern %	Percentage of use of the term for this stimulus
Global %	Overall percentage of use of the term
Intern frequency	Absolute frequency of the term for this stimulus
Global frequency	Total absolute frequency of the term
p	P-value of the test
Vtest	Standardized intensity of the association

- A positive **V-test** indicates that the term is checked more frequently for this stimulus than overall.
- A negative **V-test** indicates that it is checked less frequently — and is just as informative.
- A low **p-value** indicates a clearer characterization.

A stimulus absent from this table does not have "no profile"; it is simply close to the mean profile of all stimuli at the chosen threshold.

4.6.3 Representation of the Stimuli and the CATA *(available with 3 terms)*

This graph is produced only if at least **three** terms are placed in **CATA**. It simultaneously represents the stimuli and the terms in the factor space derived from the correspondence analysis.

- Two close stimuli have similar CATA profiles.

- A term close to a stimulus on the map is more strongly associated with that stimulus.
- The percentage displayed on each axis indicates the share of total information captured by that dimension.

This map gives an overview, but it does not replace the descriptive table. The two should be read together.

4.6.4 Representation of the Stimuli According to Clusters *(available with 3 terms)*

This graph appears only if the classification could be computed. It represents the stimuli on the same factor map, with a color code indicating which cluster each stimulus belongs to.

- Stimuli in the same cluster have similar CATA profiles.
- Distant clusters correspond to more contrasted families of stimuli.
- This output helps summarize the global structure of the stimuli.

4.6.5 Description of the Clusters *(available with 3 terms)*

When the classification is available, the function also produces a table describing each cluster through the terms that are significantly associated with it, using the same logic as for the description of stimuli.

Element	Meaning
(first column)	Group of stimuli concerned
CATA	Characteristic term
Intern %	Percentage of the term within the cluster
Global %	Overall percentage of the term
Intern frequency	Absolute frequency of the term within the cluster
Global frequency	Total absolute frequency of the term
p	P-value
Vtest	Standardized intensity of the association

This table is particularly useful for summarizing the main families of stimuli, beyond the stimulus-by-stimulus details.

4.7 Example: the *Giacalone_et_al_cata* dataset

The **Giacalone_et_al_cata** dataset describes 9 beers evaluated by 74 consumers on 15 CATA items: here, the aim is to associate beers with particular occasions. It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

4.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimulus Effect	Product
CATA	all CATA items from ThirstQ to Concert
Significance threshold (%)	5

4.7.2 Analysis of the *Description of the Stimuli According to CATA* table

The table allows a stimulus-by-stimulus reading with test values and actual frequencies.

GMA is significantly characterized by *Rugby* (V-test = 3.95, p < .001; 11.61% versus 7.08% overall) and *CampingFishing* (2.73, p = 0.006). Conversely, it is under-represented by *Gift* (V-test = -4.36), *Treat* (-3.84), and *FineDinning* (-3.51) — all highly significant.

LR has the profile most strongly marked by *Rugby* in the whole set (V-test = 5.74, p < .001; 15.24% versus 7.08% overall) and *CampingFishing* (2.83). It is under-represented by *Guests* (-4.77), *FineDinning* (-3.91), *Treat* (-3.74), and *Gift* (-3.50).

SSA stands out from all other stimuli by the intensity of its positive associations: *Gift* (V-test = 6.43, p < .001; 13.41% versus 4.21% overall), *FineDinning* (6.09), *Treat* (4.53), *Achievement* (3.71), and *Guests* (2.72). Conversely, it is strongly under-represented by *Rugby* (-4.29), *ThirstQ* (-3.63), *Work* (-3.01), *CampingFishing* (-2.88), and *TV* (-2.80). It is the stimulus whose profile differs most clearly from the mean.

SC is more strongly linked to *Rugby* (2.82) and *ThirstQ* (2.69), and is under-represented by *PubHouse* (-2.67), *Treat* (-2.58), and *Gift* (-2.55).

HPW and **PKB** have less contrasted profiles, both being more associated with *Treat* and *Gift*, and under-represented by *Rugby*.

MBB mainly stands out through *PubHouse* (2.62) and is negatively distinguished by *ThirstQ* (-2.35) and *Rugby* (-2.19).

STG is characterized only by *Rugby* on the negative side (-2.75, $p = 0.006$).

4.7.3 Analysis of the *Representation of the Stimuli and the CATA* graph

The first dimension explains 85.59% of the inertia — a remarkably high value, meaning that the main opposition between stimuli is very strong and very readable on the horizontal axis. The second dimension explains only 7.43%, but refines secondary distinctions.

On the map, *SSA* is located very clearly on the right, isolated near *Gift*, *FineDinning*, and *Achievement*. On the left are grouped *LR*, *GMA*, and *SC*, close to *Rugby* and *CampingFishing*. *MBB*, *HPW*, *PKB*, *MHR*, and *STG* occupy a central or slightly right-hand position, between the two poles. On the second dimension, *LR* is located slightly higher than *GMA* and *SC*, which reflects its stronger association with *Rugby*.

4.7.4 Analysis of the *Representation of the Stimuli According to Clusters and the Description of the Clusters* graph and table

The automatic classification retains **three clusters**:

- **Cluster 1** (black): *LR*, *GMA*, and *SC* — positively characterized by *Rugby* (V-test = 5.22, $p < .001$; 10.32% versus 7.08% overall) and *CampingFishing* (2.99), and under-represented by *Guests* (-3.71), *FineDinning* (-3.45), *Gift* (-3.43), and *Treat* (-3.17). This is the cluster of beverages associated with sports and outdoor contexts.
- **Cluster 2** (red): *MBB*, *HPW*, *PKB*, *MHR*, and *STG* — positively characterized by *Gift* (V-test = 4.06), *Guests* (2.96), *FineDinning* (2.83), and *Treat* (2.06), and under-represented by *Rugby* (-3.32) and *CampingFishing* (-2.59). This cluster groups beverages perceived as more sophisticated or associated with formal social occasions.
- **Cluster 3** (green): *SSA* alone — characterized only by *Rugby* on the negative side (-2.75). This stimulus is so singular that the classification places it alone in a separate cluster.

4.7.5 Synthetic interpretation

This example clearly shows the value of the function. It does not merely show that an item is often checked: it shows **for which stimuli it is truly characteristic**, **how the stimuli are globally opposed**, and **which main profiles** emerge from the set of responses. Here, the major opposition — captured at 85.59% by the first dimension — contrasts beverages associated with sports and outdoor contexts (*Rugby*, *CampingFishing*) with beverages perceived as more sophisticated (*Gift*, *FineDinning*, *Treat*). *SSA* is a special case, with a profile singular enough to form a cluster on its own.

What we take away from this example: CATA analysis makes it possible to move from simple checking frequencies to a structured reading of the oppositions between stimuli and the main profiles that emerge from them.

4.8 Practical tips

- Start with **Significance threshold (%) = 5%**; increase to 10% if the description tables are too empty.
- Always check the contingency table before interpreting the graphs.
- At least **three terms** are required in the **CATA** field to obtain the factor map and the classification.
- A negative **V-test** is just as informative as a positive **V-test**: it indicates that a term is checked significantly less often for a stimulus than for the others.
- Do not overinterpret the absence of a stimulus from the descriptive table: it may simply have a profile close to the mean profile.
- The classification is automatic — assess its relevance by cross-reading the graph and the cluster description.
- Read the correspondence map together with the description table: the map gives an overview, while the table provides the details.
- Make sure that the CATA columns are coherent and, most often, coded as 0/1.

4.9 In summary

This analysis transforms CATA responses into a structured reading of the sensory profiles of stimuli. It combines a **contingency table**, a **statistical description of stimuli**, then, if the number of terms is sufficient, a **correspondence analysis** and an **automatic classification**. Its main value is to move from a simple accumulation of checkmarks to a view of the stimuli that is simultaneously **descriptive**, **factorial**, and **typological**.

5 JAR Data — Defect and Penalty Analysis

5.1 Introduction

This analysis is designed for **JAR** (*Just About Right*) data, in which subjects evaluate a sensory attribute on a scale such as *too little / just about right / too much.* In this function, the analysis focuses on **defects**, that is, on all levels different from the reference JAR level, and on their relationship with the liking score. It makes it possible to describe stimuli according to their defects, to represent stimuli and defects in the same space, and then to estimate **liking penalties** associated with these defects.

Why not use a standard alternative?

A mean liking score by stimulus does not indicate which defects explain a decrease in liking. A simple reading of JAR responses also does not clearly distinguish what corresponds to a lack or an excess relative to the ideal level. Here, the function recodes each attribute around the level defined in the **Coding of the JAR level** field, constructs a stimuli $\times$ defects contingency table, and then computes liking penalties relative to the full set of stimuli.

5.2 Understanding what happens under the hood

5.2.1 Recoding of the JAR level

The function first uses the value entered in **Coding of the JAR level** to redefine, in each variable in **Sensory Attributes**, the reference level. The default value is `"jar"`. All other levels are then treated as **defects**, that is, as deviations from the level judged to be "just about right".

5.2.2 Construction of the defect frequency table

After this recoding, the function removes the reference level and counts, for each stimulus, how many times each defect was observed. The intermediate result is a **defects $\times$ stimuli** matrix, then transposed to display a **stimuli $\times$ defects** table in jamovi. This frequency table is the basis for the description of stimuli and the factorial representation.

5.2.3 Description of stimuli by defects

From this table, the function applies a frequency description procedure. For each stimulus, it identifies the defects that are significantly more or less associated than in the whole table. The logic is close to that used for CATA data: the frequency of a defect for a stimulus is compared with its global frequency, and significant associations are reported with their frequencies, p-values, and V-tests.

5.2.4 Representation of products according to defects

The function then performs a **correspondence analysis** on the defect table and produces the graph entitled **Representation of the Products According to Defects**. This map simultaneously represents the stimuli and the defects in the same factor space, in order to visualize proximities between defect profiles.

5.2.5 Calculation of penalties

The function calculates penalties from **Stimulus Effect**, **Subject Effect**, **Liking Variable**, and **Sensory Attributes**. The penalty measures the loss in liking associated with a defect: it is the difference between the mean liking score of subjects who checked the JAR level and that of subjects who checked the defect level. This difference is calculated across all stimuli simultaneously, which gives it a global interpretation, not a stimulus-specific one.

> The penalty is a negative or zero value: it represents a loss in liking. The larger the absolute value, the more penalizing the defect is.

5.2.6 A penalty graph for each stimulus

The function produces a penalty graph for **each** stimulus. The display order depends on the order of the **Stimulus Effect** levels in jamovi. It is therefore possible to place another stimulus first by modifying the order of the levels in the **Data** tab.

5.3 Expected data structure

The data must contain:

- one categorical column for **Stimulus Effect**;
- one categorical column for **Subject Effect**;
- one numeric column for **Liking Variable**;
- one or more categorical columns for **Sensory Attributes**.

Each row corresponds to the evaluation of one stimulus by one subject. The variables in **Sensory Attributes** must be factors, because the function modifies their reference level using **Coding of the JAR level**.

Before running the analysis in jamovi, check in the **Data** tab that:

- **Stimulus Effect** is categorical;
- **Subject Effect** is categorical;
- **Liking Variable** is numeric;
- all variables in **Sensory Attributes** are categorical;
- the JAR level used in your data corresponds exactly to the value entered in **Coding of the JAR level**.

5.4 Interface guide

The analysis appears in **SEDA > Fixed List of Attributes > Analysis of JAR Data**. The interface includes four variable drop zones, the **Read me before running** checkbox, and the text field **Coding of the JAR level**.

5.4.1 Variables to complete

5.4.1.1 Stimulus Effect

Role: identifies the stimuli to analyze. **Impact on the computation:** this variable defines the groups for which defect frequencies are calculated and for which the penalty graphs are generated. **Practical tip:** the order of the levels of this variable determines the display order of the penalty graphs. Modify this order in the **Data** tab if you want to see a particular stimulus first.

5.4.1.2 Subject Effect

Role: identifies the subjects. **Impact on the computation:** this variable is passed to the penalty calculation and is used to structure observations by subject. **Practical tip:** check that the same subject keeps exactly the same identifier throughout the file.

5.4.1.3 Liking Variable

Role: contains the overall liking score used to calculate the penalties. **Impact on the computation:** this is the numeric variable on which the function estimates the effect of defects. The penalty measures the loss on this score. **Practical tip:** make sure that the liking scale is consistent for all stimuli and all subjects.

5.4.1.4 Sensory Attributes

Role: groups together the JAR variables, one per sensory attribute. **Impact on the computation:** each variable is recoded around the level defined in **Coding of the JAR level**; all other levels become defects. **Practical tip:** check that each variable does contain the JAR level with the exact expected label.

5.4.2 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

5.4.3 Coding of the JAR level: Coding of the JAR level

Role: indicates the exact label of the level considered as the reference JAR level. **Impact on the computation:** this parameter determines the level removed from the defect table and used as the reference in the recoding. If the value does not match the data, the analysis may fail or produce inconsistent results. **Practical tip:** the default value is `jar`. Adapt it exactly to the label used in your file, respecting case.

5.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical, the subject variable is categorical, the liking variable is numeric, and all JAR variables are categorical.
3. Note the exact label of the "just about right" level used in your JAR variables.
4. Open **SEDA > Fixed List of Attributes > Analysis of JAR Data**.
5. Place the stimulus variable in **Stimulus Effect**.
6. Place the subject variable in **Subject Effect**.
7. Place the liking variable in **Liking Variable**.
8. Place all JAR variables in **Sensory Attributes**.

9. Enter in **Coding of the JAR level** the exact label of the JAR level; leave `jar` if this is indeed the label used in your data.
10. First read the defect frequency table to check the data structure.
11. Then consult the description of stimuli according to defects.
12. Examine the **Representation of the Products According to Defects** map for an overview.
13. Read the penalty table, then the **Penalties for Product ...** graphs to identify priority defects.

5.6 Interpretation of results

5.6.1 Frequency Table — defect frequency table

This table crosses stimuli in rows and non-JAR levels in columns. Each cell indicates the number of times a defect was observed for a given stimulus. It is the basis for all subsequent outputs.

Element	Meaning
High value in a cell	The defect is frequent for this stimulus
Row with several high values	The stimulus accumulates several frequent defects
Column high for several stimuli	Frequent defect, but not necessarily specific

> **Point of vigilance:** a very rare defect will contribute little to the interpretation. A column with very low frequencies generally indicates a weakly discriminating defect.

5.6.2 Description of the Stimuli According to Defects

This table describes, stimulus by stimulus, the significantly associated defects.

Element	Meaning
(first column)	Stimulus concerned
Defect	Characteristic defect
Intern %r	Percentage of the defect for this stimulus
Global %	Global percentage of the defect
Intern frequency	Absolute frequency of the defect for this stimulus
Global frequency	Total absolute frequency of the defect
p	P-value of the test

Element	Meaning
Vtest	Standardized intensity of the association

- A positive **V-test** indicates that the defect is over-represented for this stimulus.
- A negative **V-test** indicates that it is under-represented — which is also informative.
- A low **p-value** indicates a clearer association.

  A stimulus absent from this table has no defect significantly associated at the threshold used: its defect profile is close to the mean of the full set of stimuli.

5.6.3 Representation of the Products According to Defects — graph

This graph comes from the correspondence analysis on the defect table. It simultaneously represents the stimuli and the defects in the same factor space. Two close stimuli have similar defect profiles; a defect close to a stimulus is more strongly associated with it. The percentage displayed on each axis indicates the share of information captured by that dimension.

This map gives an overview and should be complemented by reading the description table for the statistical details.

5.6.4 Penalty Table — penalty table

This table presents, for each defect, the estimated effect on the liking score.

Element	Meaning
Defect	Defect concerned
Penalty	Estimated loss in liking associated with this defect
Std. Error	Standard error of the estimate
p	P-value of the test

- The larger the absolute value of **Penalty**, the stronger the estimated effect.
- A low **p-value** suggests an effect more clearly supported by the data.
- A weakly significant penalty does not necessarily imply the absence of an effect: the defect may simply be too rare to be estimated precisely.

5.6.5 Penalties for Product ... — graphs by stimulus

The function generates a penalty graph for each stimulus. The x-axis corresponds to the frequency of the defect for this stimulus; the y-axis corresponds to the penalty calculated across all stimuli. Error bars are drawn around the points, defects with $p < 0.05$ are visually distinguished, and labels appear for defects with $p < 0.20$.

Element	Meaning
High frequency + marked penalty	Priority defect: frequent and strongly penalizing
High frequency + weak penalty	Frequent but less penalizing defect
Low frequency + marked penalty	Rare but potentially problematic defect
Low frequency + weak penalty	Secondary defect

> **Point of vigilance:** the graph crosses **local** information (the frequency of the defect for the displayed stimulus) with **global** information (the penalty calculated across all stimuli). This point is essential to avoid overinterpreting an isolated point on the graph.

5.7 Example: the milkshake dataset

The **milkshake** dataset concerns three milkshakes evaluated using several JAR attributes related in particular to odor, texture, fluidity, sweetness, and acidity. It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

5.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimulus Effect	Product
Subject Effect	Panelist
Liking Variable	Liking
Sensory Attributes	all JAR variables from Color to Taste_Van
Coding of the JAR level	`jar`

5.7.2 Analysis of the *Frequency Table (Stimuli × Defects)*

The frequency table shows that the three products have very different defect profiles.

The **859** milkshake is strongly marked by *Not_enough_texture*, mentioned by 38 subjects out of 47 (80.85%), and *Too_fluid* (70.21%); these are the two most frequent defects in the whole set.

The **972** milkshake stands out mainly through *Not_enough_odor_int* (53.19%), *Not_enough_odor_fruit* (46.81%), *Not_enough_odor_van* (44.68%), and *Too_much_sweetness* (42.55%).

The **284** milkshake has several frequent but less extreme defects: *Not_enough_texture* (57.45%), *Not_enough_odor_int* (48.94%), *Too_fluid* (46.81%), and *Not_enough_odor_van* (44.68%).

This first reading is important: it makes it possible to identify which defects dominate sensorially for each product, even before knowing whether they are truly associated with a decrease in liking.

5.7.3 Analysis of the *Description of the Products According to Defects* table

This table identifies the defects that are significantly over- or under-represented for each product relative to the *average product*.

The **859** milkshake is the one whose profile stands out most clearly. It is more strongly associated with *Not_enough_texture* (p = 0.055, V-test = 1.92) and *Too_fluid* (p = 0.080, V-test = 1.75). Conversely, it is significantly less associated with *A_little_plain* (p = 0.040, V-test = -2.05) and *Not_enough_odor_van* (p = 0.049, V-test = -1.97). Its sensory profile is therefore mainly driven by texture and fluidity defects, rather than by a lack of vanilla odor.

The **972** milkshake is more strongly associated with *Too_much_sweetness* (p = 0.087, V-test = 1.71), while the other positive associations are far from significant at the 5% threshold. Conversely, it is less associated with *Not_enough_texture* (p = 0.072, V-test = -1.80) and *Not_enough_sweetness* (p = 0.099, V-test = -1.65), which places it in partial opposition to the **859** milkshake on these dimensions.

The **284** milkshake appears much closer to the mean profile. No defect stands out clearly at the 5% threshold, suggesting a product less singular than the other two from the point of view of its defects.

5.7.4 Analysis of the *Representation of the Products According to Defects* graph

The first dimension explains 74.78% of the inertia and clearly opposes the **859** milkshake to the **972** milkshake. On the **859** side, the main defects are *Too_fluid*, *Not_enough_texture*, and *Not_enough_sweetness*. On the **972** side, the defects are rather *Too_much_sweetness*, *Too_much_odor_van*, and *Not_enough_odor_fruit*.

The second dimension (25.22%) may be more specific to the **284** milkshake, which is associated with defects related to fruity or vanilla odors.

> As an essential complement to the tables, this graph, derived from a correspondence analysis, represents the associations between products and defects, making proximities and oppositions between products immediately visible.

5.7.5 Analysis of the *Penalties Obtained from All Defects* table

The penalty table provides a **global** reading of the impact of defects on liking, across all products. It does not say how frequently a defect appears for a given product; it indicates how much this defect is associated with an average loss in liking when it is observed.

In this example, only one defect is significant at the 5% threshold: *Too_much_odor_fruit*, with a penalty of -2.67 and a p-value of 0.018. It is therefore the only defect whose negative effect on liking is clearly supported by the data. Several other defects show notable penalties without reaching this significance threshold, for example *Too_much_acidity* (-4.03, p = 0.082), *Too_much_texture* (-1.78, p = 0.089), and *Not_enough_acidity* (-1.03, p = 0.087). They cannot be considered statistically significant here, but they are weak signals to monitor.

> **Point of vigilance:** penalties are calculated across all products simultaneously. They measure **global** information — which defect penalizes liking the most, across all products — and not **local** information, such as how frequently this defect appears for a given product. It is precisely the combination of these two pieces of information that will be useful.

5.7.6 Analysis of the *Representation of the Penalties for Each Product* graphs

These three graphs cross two essential pieces of information for product optimization:

- on the **x-axis**, the **frequency** of the defect for the displayed product, derived from the *Frequencies of Defects by Product and Sensory Attribute* table;
- on the **y-axis**, the **penalty** associated with this defect, calculated across the full panel and all products, derived from the *Penalties Obtained from All Defects* table.

By construction, a defect therefore keeps the same y-coordinate from one graph to another; only its x-coordinate varies according to its prevalence in the product considered. Red points indicate defects whose penalty is statistically significant at the 5% threshold. In our example, only *Too_much_odor_fruit* stands out through a significant penalty.

To be operational, it is essential to combine this reading with the *Description of the Products According to Defects* table, which identifies the **specificity** of each defect (over- or under-representation). A frequent defect is not necessarily specific, and conversely, strong specificity does not systematically imply a strong penalty on liking. Taking these three pieces of information into account (local frequency, global penalty, specificity) makes it possible to prioritize improvement levers.

For the **284** milkshake, no defect stands out significantly in the descriptive table (profile close to the mean). The graph shows that the critical defect *Too_much_odor_fruit* is very rare (6.38%). As for *Too_much_acidity*, although it is strongly penalizing (-4.03), its complete absence for this product (0.00%) disqualifies it as an improvement avenue. Lacks of vanilla or fruity odor are frequent, but without being specific to the product or penalizing. The **284** milkshake therefore presents no priority optimization lever.

The analysis of the **859** milkshake shows a clear convergence between a tendency toward over-representation (V-test > 1.7, without significance at the 5% threshold) and high prevalence for texture defects (*Too_fluid* at 70.21% and *Not_enough_texture* at 80.85%). Although their global hedonic cost is moderate, they are the clearest negative traits of this product. Moreover, the critical defect *Too_much_odor_fruit* remains marginal (4.26%). The main issue for **859** therefore lies in correcting its **fluidity** and **texture**, rather than adjusting its aroma.

The **972** milkshake presents a different configuration. Its most over-represented defect is **Too_much_sweetness** (V-test = 1.71, p = 0.087), which is relatively frequent (42.55%). However, since its global penalty is low, it does not represent a major optimization lever. As for the other products, the critical defect *Too_much_odor_fruit* is almost absent (4.26%). The other aromatic lacks are frequent but have little impact. The **972** milkshake therefore does not show any defect combining high frequency, specificity, and marked penalty.

5.7.7 Synthetic interpretation

In summary, these graphs clearly show that the **penalty** provides information about the **hedonic cost** of a defect, while its **frequency** indicates its actual weight for a given product. To turn this reading into an optimization avenue, however, it must be crossed with the **specificity** of the defect, as shown in *Description of the Products According to Defects*. This threefold reading leads here to three distinct diagnoses: the **284** milkshake appears close to the mean profile, with no clearly identified priority lever; the **859** milkshake is mainly distinguished by **texture** and **fluidity** defects, both specific and very frequent; the **972** milkshake presents an excess of **sweetness** that is more specific than truly penalizing.

What we take away from this example: JAR analysis is not only used to identify the most frequent defects, but to distinguish those that are **frequent**, those that are **specific to the product**, those that are **penalizing**, and above all those that combine several of these dimensions. It is this cross-reading that makes it possible to prioritize product optimization actions.

5.8 Practical tips

- Always check that **Coding of the JAR level** corresponds exactly to the label used in your data, including case.
- Make sure that all JAR variables are categorical in jamovi before running the analysis.
- Always read the frequency table before interpreting penalties: a high penalty on a very rare defect is difficult to use in product optimization.
- Focus first on defects that are both frequent and associated with a marked and significant penalty.
- Do not overinterpret a weakly significant penalty as a complete absence of effect.
- Keep in mind that the penalty is calculated globally across all stimuli, whereas the frequency displayed in the graph is specific to the displayed stimulus.
- Use the **Representation of the Products According to Defects** map for the overview, and the tables for the details.
- If you want to display another stimulus first in the penalty graphs, modify the order of the **Stimulus Effect** levels in the **Data** tab.

5.9 In summary

This analysis transforms JAR responses into a structured reading of perceived defects and their relationship with liking. It combines a **defect frequency table**, a **description of stimuli according to defects**, a **factorial representation of products and defects**, and then an **estimation of the penalties** associated with each defect. Its main value is to connect, within the same framework, the frequency of deviations from the ideal level (jar) and their impact on liking, in order to guide product optimization.

6 Napping Data — Consensus Representation

6.1 Introduction

This analysis is designed for **Napping** (*projective mapping*) data, in which each subject positions a set of stimuli in a two-dimensional space according to their perceived similarities and differences. It makes it possible to construct a **consensus representation** of the stimuli from several individual maps and, if desired, to group the stimuli into **clusters**. The function relies on a **Multiple Factor Analysis (MFA)** to merge the different pairs of coordinates into a single space.

Why not use a standard alternative?

A PCA applied directly to all coordinate columns would treat each variable as independent, whereas in Napping the data are naturally organized by **X/Y pairs** provided by each subject. Here, each pair is treated as a **group** in the MFA, which makes it possible to construct a more suitable compromise between the different individual maps. In addition, this MFA is performed **without standardizing the data**: by positioning the stimuli on a rectangle with fixed dimensions, each subject implicitly expresses that their first dimension of variability is more important than the second. Standardizing the data would erase this fundamental information.

6.2 Understanding what happens under the hood

6.2.1 The data are read as pairs of coordinates

The function expects in **(X,Y) Coordinates** a set of numeric columns organized in pairs: X then Y for a first subject, X then Y for a second subject, and so on. The same principle applies to **Supplementary (X,Y) Coordinates**. If the number of columns is odd in either of the two blocks, the analysis is rejected. In practice, the main block must contain at least **4 columns**, that is, **2 pairs**.

6.2.2 An MFA on groups of size 2, without standardization

The core of the analysis is an **MFA**. Each X/Y pair in **(X,Y) Coordinates** is treated as a group of two quantitative variables. The MFA constructs a common space that summarizes the information carried by all subjects, while **balancing the contribution of each group**: a subject who differentiated the stimuli only slightly will not have more influence than a subject who discriminated them very clearly. The data are not standardized, which preserves the relative importance of the two axes of the rectangle.

6.2.3 The role of supplementary coordinates

If you provide variables in **Supplementary (X,Y) Coordinates**, they are projected as **supplementary groups** in the MFA. They do not participate in the construction of the axes, but make it possible to locate other subjects in the same space.

6.2.4 Automatic description of the dimensions

Once the MFA has been computed, the function produces an **automatic description of the dimensions** filtered by **Significance threshold (%)** and **Number of dimensions**. This output helps understand which variables are most strongly linked to the first axes.

6.2.5 Saving components and clustering

The **Number of components** field determines the number of components calculated and saved in **Coordinates**. If you activate **Representation of the clusters**, the function runs an **HCA** on the result of the MFA. By default, clustering is based on the **first 5 components**, and **Number of clusters = -1** means that the number of clusters is chosen automatically.

6.3 Expected data structure

The data must contain:

- one optional but strongly recommended column for **Stimuli Labels**;
- at least **two pairs** of numeric columns in **(X,Y) Coordinates**;
- optionally, numeric pairs in **Supplementary (X,Y) Coordinates**.

Each row corresponds to a **stimulus**. This distinguishes this analysis from other analyses where each row corresponds to an individual response. If **Stimuli Labels** is not completed, the function uses row numbers as stimulus names.

Before running the analysis in jamovi, check in the **Data** tab that:

- **Stimuli Labels** is categorical if you use it;
- all columns in **(X,Y) Coordinates** are numeric;
- all columns in **Supplementary (X,Y) Coordinates** are numeric;
- the number of columns in each of the two coordinate blocks is **even**.

6.4 Interface guide

The analysis appears in **SEDA > Analysis of Napping Data**. The interface includes three variable drop zones, followed by the **Graphic Options**, **Numerical Indicators**, and **Clustering** sections.

6.4.1 Variables to complete

6.4.1.1 Stimuli Labels

Role: provides the names of the stimuli in the outputs. **Impact on the computation:** this variable does not modify the MFA itself, but determines the labels displayed in the graphs and the names associated with the saved coordinates. **Practical tip:** use short and unique labels; otherwise, the maps quickly become difficult to read.

6.4.1.2 (X,Y) Coordinates

Role: groups together the active coordinates used to construct the consensus representation. **Impact on the computation:** each X/Y pair becomes a group in the MFA. The number of columns must be even, and in practice the function requires at least **4 columns**. **Practical tip:** strictly respect the order X, Y, X, Y… to avoid an incoherent map.

6.4.1.3 Supplementary (X,Y) Coordinates

Role: adds pairs of coordinates projected as supplementary groups. **Impact on the computation:** these coordinates do not influence the construction of the axes, but are projected into the MFA space. **Practical tip:** use this block to project a second panel or additional information without modifying the main compromise.

6.4.2 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

6.4.3 Graphic Options

6.4.3.1 X-axis

Role: chooses the dimension displayed on the horizontal axis. **Impact on the computation:** it does not modify the MFA, but changes the factor plane displayed in the graphs. **Practical tip:** start with 1, the default value.

6.4.3.2 Y-axis

Role: chooses the dimension displayed on the vertical axis. **Impact on the computation:** it does not modify the MFA, but changes the factor plane displayed in the graphs. **Practical tip:** start with 2, the default value.

6.4.4 Numerical Indicators

6.4.4.1 Significance threshold (%)

Role: sets the significance threshold for the automatic description of the dimensions. **Impact on the computation:** the stricter the threshold, the more selective the automatic description will be. **Practical tip:** the default value of 5% is a good starting point.

6.4.4.2 Number of dimensions

Role: determines the number of axes described automatically. **Impact on the computation:** it controls the dimension description output and also influences the minimum number of components calculated. **Practical tip:** leave 2 for a first reading.

6.4.4.3 Number of components

Role: defines the number of components calculated and saved. **Impact on the computation:** it directly influences the coordinates saved in **Coordinates** and serves as the basis for clustering. **Practical tip:** the default value of 5 is consistent with the clustering proposed by the function.

6.4.4.4 Coordinates

Role: saves the factorial coordinates of the stimuli in the jamovi data sheet. **Impact on the computation:** it does not add a new calculation, but saves the results of the MFA. **Practical tip:** activate this output if you want to reuse the dimensions in other jamovi analyses.

6.4.5 Clustering

6.4.5.1 Representation of the clusters

Role: activates the display of the clustering graph. **Impact on the computation:** when this option is checked, the function computes the HCA if necessary and displays the map of stimuli according to clusters. **Practical tip:** leave it unchecked for a first reading, then activate it to complement the consensus map with a group-based reading.

6.4.5.2 Number of clusters

Role: sets the number of clusters to extract. **Impact on the computation:** with -1, the number of clusters is determined automatically; any positive value imposes that number. **Practical tip:** start with -1, then set a number only if you have a clear analytical justification.

6.4.5.3 Cluster variable

Role: saves the cluster membership of each stimulus as a new variable in jamovi. **Impact on the computation:** it does not add a new calculation, but saves the results of the classification. **Practical tip:** useful if you then want to cross the clusters with other variables in jamovi.

6.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the coordinate columns are numeric and organized in X/Y pairs.
3. If you have a stimulus identifier column, check that it is categorical.
4. Open **SEDA > Analysis of Napping Data**.
5. Place the stimulus name column in **Stimuli Labels**.
6. Place all active pairs in **(X,Y) Coordinates**.
7. If needed, place the supplementary pairs in **Supplementary (X,Y) Coordinates**.
8. Check that the number of columns in each coordinate block is even.

9. Leave **X-axis = 1** and **Y-axis = 2**.
10. Leave **Significance threshold (%) = 5**, **Number of dimensions = 2**, and **Number of components = 5**.
11. First read **Representation of the Stimuli**, then **Representation of the Subjects**, then the eigenvalue table.
12. Then consult **Automatic Description of the Dimensions**.
13. Activate **Representation of the clusters** if you want to visualize the classification results in **Representation of the Stimuli According to Clusters**.
14. Optionally save **Coordinates** and **Cluster variable** if you want to reuse the results.

6.6 Interpretation of results

6.6.1 Representation of the Stimuli

This is the important result of the function. It represents the stimuli in the consensus factor space constructed by the MFA. Two close stimuli were positioned similarly by the set of subjects; two distant stimuli are perceived as more different. The percentage displayed on each axis indicates the share of variance explained by the corresponding dimension.

Element	Meaning
Two close stimuli	Similar global perceptions
Two distant stimuli	Different global perceptions
Compact group of stimuli	Family of stimuli perceived as close

> **Point of vigilance:** the stimulus map shows a perceptual structure, but it is not always sufficient to name the axes. Adding supplementary information, such as verbalization, can greatly help interpretation.

6.6.2 Representation of the Subjects

This map represents the X/Y coordinate groups — that is, the subjects or individual maps — in the factor space. It helps assess the level of consensus between subjects.

Element	Meaning
Close subjects	Globally coherent individual maps
Dispersed subjects	Weaker consensus
Subject close to the origin	Perception less aligned with the dominant axes

6.6.3 Eigenvalue Decomposition — eigenvalue table

Element	Meaning
Dim.	Dimension number
Eigenvalue	Importance of the dimension
% of variance	Share of variability explained
Cumulative %	Cumulative share explained

If the first two dimensions explain a large share of the variance, the 1/2 plane provides a good summary; otherwise, it may be useful to explore other factor planes.

6.6.4 Automatic Description of the Dimensions

This output automatically describes the retained dimensions according to **Significance threshold (%)** and **Number of dimensions**. It helps identify which variables are most strongly linked to the axes and becomes particularly useful when **Supplementary (X,Y) Coordinates** have been added.

6.6.5 Representation of the Stimuli According to Clusters *(available if "Representation of the clusters" is activated or if "Cluster variable" is requested)*

This graph shows the stimuli in the factor space, grouped according to the clusters derived from the HCA.

Element	Meaning
Stimuli in the same cluster	Similar positioning profiles
Clearly separated clusters	Well-differentiated families of stimuli

6.6.6 Coordinates *(data output)*

This output saves the factorial coordinates of the stimuli on the first calculated dimensions. Each column corresponds to one dimension (`Dim. 1`, `Dim. 2`, etc.).

6.6.7 Cluster variable *(data output)*

This output saves the cluster membership of each stimulus as a nominal variable. It is useful for subsequent analyses in jamovi.

6.7 Example: the perfumes_napping dataset

The **perfumes_napping** dataset describes 12 perfumes positioned on a sheet (with dimensions 60 by 40) by 60 subjects, that is, 120 coordinate columns (one X/Y pair per subject). It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

6.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimuli Labels	variable `Ident`
(X,Y) Coordinates	the first 60 coordinate columns (30 pairs)
Supplementary (X,Y) Coordinates	the following 60 coordinate columns (30 pairs)
X-axis / Y-axis	1 / 2
Significance threshold (%)	5
Number of dimensions	2
Number of components	5
Representation of the clusters	checked to examine the classification
Number of clusters	-1

6.7.2 Analysis of the *Representation of the Stimuli* graph

The map reveals a fairly clear perceptual structure on the Dim. 1 (25.07%) / Dim. 2 (18.03%) plane. On the horizontal axis, **Aromatics Elixir** and **Shalimar** are clearly located on the right (positive coordinates on Dim. 1), whereas **J'adore EP**, **J'adore ET**, **Coco Mademoiselle**, **Cinéma**, **Pleasures**, and **Pure Poison** are located on the left. **Chanel N5** differs from this right-hand group by a low position on the second dimension (strongly negative value on Dim. 2), which slightly isolates it from the Shalimar / Aromatics Elixir pair. On the vertical axis, **Angel**, **Lolita Lempicka**, and **L'instant** are clearly located above the center, forming a coherent perceptual group at the top of the plane.

6.7.3 Analysis of the *Representation of the Subjects* graph

The subject map represents each subject (S1 to S60) by their coordinates on the two dimensions. Subjects whose coordinates on Dim. 1 are high (**S1**, **S20**, **S16**, **S57**, **S60**) strongly structured their rectangle around the main opposition — the one separating Aromatics Elixir / Shalimar from the left-hand group. Conversely, subjects such as **S4**, **S2**, **S34**, or **S33** have higher coordinates on Dim. 2, contributing more to the vertical structure that isolates Angel and Lolita Lempicka. Subjects close to the origin, many of whom are located at the center of the map, contribute more moderately to one or the other dimension.

6.7.4 Analysis of the *Eigenvalue Decomposition* table

The first two dimensions explain 43.09% of the total variance (Dim. 1: 25.07%, Dim. 2: 18.03%). In Napping, because individual rectangles differ greatly from one subject to another, this concentration of information is rather good for two dimensions. The following dimensions provide decreasing but non-negligible contributions (Dim. 3 : 12.31%, Dim. 4: 9.58%, Dim. 5: 7.96%) — which means that other factor planes could reveal additional oppositions not visible here.

6.7.5 Analysis of the *Automatic Description of the Dimensions* table

The **Automatic Description of the Dimensions** output identifies the individual coordinates (X and Y for each subject) most correlated with each axis. For **Dim. 1**, the strongest positive correlations are **X20** (r = 0.9254), **X1** (r = 0.9187), and **Y21** (r = 0.9003): these subjects positioned the right-hand perfumes at the far right of their rectangle. The strongest negative correlations are **Y48** (r = -0.9126) and **X42** (r = -0.8539): these subjects, conversely, placed the left-hand perfumes at the far right of their own rectangle. This coherence between very different subjects confirms that Dim. 1 captures a strong and shared perceptual opposition, even if not all subjects express it with the same intensity.

6.7.6 Analysis of the *Representation of the Stimuli According to Clusters* graph

With **Number of clusters = -1**, the automatic classification retains **three clusters**:

- **Cluster 1** (black): **Coco Mademoiselle**, **J'adore EP**, **J'adore ET**, **Cinéma**, **Pleasures**, and **Pure Poison** — perfumes located on the left of the plane, generally perceived as lighter or floral;
- **Cluster 2** (red): **Angel**, **Lolita Lempicka**, and **L'instant** — perfumes located at the top of the plane, perceived as forming a distinct family on the second dimension;
- **Cluster 3** (green): **Shalimar**, **Aromatics Elixir**, and **Chanel N5** — perfumes located on the right and lower part of the plane, perceived as stronger or more intense.

6.7.7 Synthetic interpretation

This example clearly shows the value of the function. It does not only provide an average map of the perfumes: it also shows how the perfumes are distributed in a **consensus space**, to what extent the **subjects converge** on this structure, and which **groups of perfumes** emerge from the classification. The global structure clearly opposes three perceptual families, even though the first two dimensions capture only 43% of the total variability.

> **What we take away from this example:** Napping analysis makes it possible to transform a set of individual maps into a readable consensus representation, while showing the level of convergence between subjects and the main groups of stimuli.

6.8 Practical tips

- Always check that the columns in **(X,Y) Coordinates** are provided in X/Y pairs and that their total number is even.
- Use **Stimuli Labels** almost systematically to avoid reading the map with simple row numbers.
- Start with **X-axis = 1** and **Y-axis = 2**; if the first two dimensions summarize little variance, explore other factor planes.
- Keep **Number of components = 5** for a first trial, especially if you are considering clustering.
- Leave **Number of clusters = -1** at first to let the function choose automatically.
- Do not read the stimulus map without also looking at **Representation of the Subjects**: it helps assess consensus between subjects.
- The stimulus map does not speak for itself: use **Supplementary (X,Y) Coordinates** or verbalization data to give sensory meaning to the axes.
- Activate **Coordinates** and **Cluster variable** if you want to reuse the results in other jamovi analyses.

6.9 In summary

This analysis constructs a **consensus map of the stimuli** from Napping data, by combining the individual maps through a non-standardized **MFA** that balances the contribution of each subject. It simultaneously represents the structure of the stimuli and the level of consensus between subjects, and can be complemented by an **automatic clustering** of the stimuli. Its main value is to transform a set of individual maps into a common and directly interpretable representation of perceptual proximities, without requiring a predefined list of attributes.

7 Sorting Data — Analysis of Categorization

7.1 Introduction

This analysis is designed for **free sorting** (*sorting task*) data, in which each subject groups stimuli into categories according to their own perception. It makes it possible to obtain a **multivariate representation of the stimuli** from the categorizations provided by the subjects, and, if needed, an **automatic classification** of the stimuli into groups. When the groups have been verbalized, the function also provides a **textual description of the stimuli** based on the words used to name these groups.

Why not use a standard alternative?

A classical approach consists of transforming the sorting data into a similarity matrix between stimuli, then applying a projection method such as MDS. This strategy is useful, but it removes part of the individual information carried by each subject. Here, the function directly analyzes the columns in **Subjects** as qualitative variables and applies a **Multiple Correspondence Analysis (MCA)**, which preserves more of the structure of individual responses and makes it possible to represent stimuli and categories simultaneously in the same space.

7.2 Understanding what happens under the hood

7.2.1 The stimuli are the individuals in the analysis

The function reads the columns placed in **Subjects** as categorical variables. Each row corresponds to a **stimulus**, and each column indicates in which group a subject placed this stimulus. If **Stimuli Labels** is completed, this variable is used to name the stimuli in the outputs; otherwise, the function uses row numbers.

7.2.2 An MCA on the categorizations provided by the subjects

The core of the function is an **MCA** applied to the columns in **Subjects**. In this framework, two stimuli are close if they have often been grouped in a similar way by the subjects. The columns placed in **Supplementary Subjects**, if any, are projected as supplementary qualitative variables: they appear in the factor space, but do not participate in the construction of the axes.

7.2.3 The role of Ventilation level (%)

The **Ventilation level (%)** parameter reduces the effect of very infrequent categories in the factorial results. In free sorting, some subjects may isolate a stimulus in a single-item group, creating a very rare category that would otherwise have a disproportionate influence on the MCA. This threshold limits this leverage effect. The default value is 5%.

7.2.4 Textual description of the stimuli

The function constructs a stimuli × words correspondence table from the group labels, then applies a frequency description procedure. This part is used to identify, for each stimulus, the most characteristic words among the labels used by the subjects. It becomes particularly useful when the groups have been named with interpretable sensory terms.

7.2.5 Automatic description of the dimensions and clustering

Once the MCA has been computed, the function produces an **automatic description of the dimensions** according to **Significance threshold (%)** and **Number of dimensions**. If you activate **Representation of the clusters**, the function applies an **HCA** to the MCA components. By default, clustering is based on the **first 5 components**, and **Number of clusters = -1** lets the function choose the number of groups automatically.

7.3 Expected data structure

The data must contain:

- one optional column for **Stimuli Labels**;
- at least **two** categorical columns in **Subjects**;
- optionally, categorical columns in **Supplementary Subjects**.

Each row corresponds to a **stimulus**. This point is important: unlike in other analyses, here a row does not correspond to an individual response, but to a stimulus described by the categorizations provided by several subjects.

Before running the analysis in jamovi, check in the **Data** tab that:

- **Stimuli Labels** is categorical if you use it;

- all columns in **Subjects** are categorical (factors);
- all columns in **Supplementary Subjects** are categorical;
- you have placed at least **two** variables in **Subjects**.

7.4 Interface guide

The analysis appears in **SEDA > Analysis of Sorting Data**. The interface includes three variable drop zones, a **Read me before running** checkbox, and then the **Random Assignment for Rare Categories**, **Graphic Options**, **Numerical Indicators**, and **Clustering** sections.

7.4.1 Variables to complete

7.4.1.1 Stimuli Labels

Role: provides the names of the stimuli in the outputs. **Impact on the computation:** it does not modify the MCA, but determines the labels displayed in the graphs and the names associated with the saved coordinates. **Practical tip:** use short and unique labels; otherwise, the maps become difficult to read.

7.4.1.2 Subjects

Role: groups together the sorting variables, one per subject. **Impact on the computation:** these are the active variables of the MCA. They are also used for the textual description and clustering. At least **two** variables are required. **Practical tip:** check that each column is categorical in jamovi.

7.4.1.3 Supplementary Subjects

Role: groups together variables projected as supplementary qualitative variables. **Impact on the computation:** they do not influence the construction of the axes, but their position in the space can help interpretation. **Practical tip:** useful for projecting a second group of subjects without modifying the main result.

7.4.2 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

7.4.3 Random Assignment for Rare Categories: Ventilation level (%)

Role: sets the threshold below which a category is considered rare and treated in a way that reduces its influence. **Impact on the computation:** categories whose relative frequency is below this threshold have a reduced weight in the MCA. The default value is 5%. **Practical tip:** leave 5% for a first trial. If some very rare categories seem to dominate the map, you can test a slightly higher threshold.

7.4.4 Graphic Options *(collapsed by default)*

7.4.4.1 X-axis and Y-axis

Role: choose the dimensions displayed on the horizontal and vertical axes. **Impact on the computation:** they do not modify the MCA, but change the displayed factor plane. **Practical tip:** start with **X-axis = 1** and **Y-axis = 2**, the default values.

7.4.4.2 Active categories

Role: controls the display of active categories on the category map. **Impact on the computation:** no effect on the MCA; it only affects visibility in the graph. **Practical tip:** leave it checked at first. Uncheck it if the map is too crowded.

7.4.4.3 Selection of categories

Role: filters the displayed categories according to a quality-of-representation criterion. **Impact on the computation:** no effect on the MCA; it only affects display. **Practical tip:** the default value **cos2 10** is a good compromise for lightening the map without losing the best-represented categories.

7.4.4.4 Supplementary categories

Role: controls the display of categories derived from **Supplementary Subjects**. **Impact on the computation:** no effect on the MCA. **Practical tip:** leave it checked if you use supplementary variables and want to compare them visually with the active categories.

7.4.5 Numerical Indicators *(collapsed by default)*

7.4.5.1 Significance threshold (%)

Role: sets the significance threshold for the automatic description of the dimensions. **Impact on the computation:** the stricter the threshold, the more selective the output. **Practical tip:** the default value of 5% is well suited to a first reading.

7.4.5.2 Number of dimensions

Role: determines the number of axes described automatically. **Impact on the computation:** it controls the dimension description output. **Practical tip:** leave 2 for a first trial.

7.4.5.3 Level description (%)

Role: sets the threshold used for the textual description of the stimuli. **Impact on the computation:** a stricter threshold produces a shorter description; a broader threshold retains more words. **Practical tip:** leave 5% at first. If the textual description is too sparse, test a slightly broader threshold.

7.4.5.4 Long format dataset

Role: displays the intermediate textual dataset in long format. **Impact on the computation:** it does not add a new calculation; it makes it possible to check how the labels were reorganized before the textual analysis. **Practical tip:** activate this option if you want to inspect the transformation of the verbal data.

7.4.5.5 Individual Table Options — Coordinates, Contributions, Cosine

Role: activate the display of the numerical tables of coordinates, contributions, and cosines for the stimuli. **Impact on the computation:** no effect on the MCA; they add numerical result tables. **Practical tip:** disabled by default. Activate them if you want precise values for complementary analyses.

7.4.5.6 Variable Table Options — Coordinates, Contributions, Cosine

Role: same principle for the MCA categories. **Impact on the computation:** no effect on the MCA. **Practical tip:** useful if you want to relate the categories precisely to the factor map.

7.4.5.7 Number of components *(Save)*

Role: defines the number of components calculated and saved in **Coordinates**, and used for the classification. **Impact on the computation:** clustering computes distances on these components. **Practical tip:** the default value of 5 is consistent with the clustering proposed by the function.

7.4.5.8 Coordinates *(Save)*

Role: saves the factorial coordinates of the stimuli in the jamovi data sheet. **Impact on the computation:** no effect on the MCA. **Practical tip:** activate it if you want to reuse the components in other jamovi analyses.

7.4.6 Clustering *(collapsed by default)*

7.4.6.1 Representation of the clusters

Role: activates the computation of the hierarchical classification and the display of the corresponding graph. **Impact on the computation:** triggers an HCA on the MCA components. It is unchecked by default. **Practical tip:** leave it unchecked for a first reading, then activate it to complement the factor map with a group-based reading.

7.4.6.2 Number of clusters

Role: sets the number of clusters to extract. **Impact on the computation:** with -1, the number is chosen automatically; any positive value imposes that number. **Practical tip:** start with -1, then set a number only if you have a clear analytical reason.

7.4.6.3 Cluster variable *(Save)*

Role: saves the cluster membership of each stimulus in the jamovi data sheet. **Impact on the computation:** it does not add a new calculation, but saves the classification results. **Practical tip:** useful if you then want to cross the clusters with other variables in jamovi.

7.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that all columns in **Subjects** are categorical and that you have at least two of them.
3. Check that **Stimuli Labels** is categorical if you use it.
4. Open **SEDA > Analysis of Sorting Data**.
5. Place the stimulus name variable in **Stimuli Labels**.
6. Place the sorting variables in **Subjects**.
7. If needed, place other sorting variables in **Supplementary Subjects**.
8. Leave **Ventilation level (%) = 5**, **X-axis = 1**, **Y-axis = 2**, **Significance threshold (%) = 5**, **Level description (%) = 5**, **Number of dimensions = 2**, and **Number of components = 5** for a first trial.
9. If subjects verbalized their groups, first read **Description of the Stimuli**, then **Representation of the Individuals**, then **Representation of the Categories**.
10. Consult **Automatic Description of the Dimensions** to interpret the axes.
11. Read the eigenvalue table to assess whether the 1/2 plane summarizes the data well.
12. Activate **Representation of the clusters** if you want an automatic classification.

7.6 Interpretation of results

7.6.1 Description of the Stimuli

This table describes each stimulus based on the words used in the group labels. It is truly informative only if the groups were named with interpretable words.

Element	Meaning
First column	Stimulus concerned
Word	Characteristic word
Inter %	Percentage of the word for this stimulus
Global %	Global percentage of the word
Intern frequency	Frequency of the word for this stimulus
Global frequency	Global frequency of the word
p	P-value of the test
Vtest	Standardized intensity of the association

- A word with an **Intern %** clearly higher than **Global %r** characterizes this stimulus more strongly.
- A low **p-value** indicates a clearer association.
- A positive **V-test** indicates over-representation of the word for this stimulus; a negative **V-test** indicates under-representation.

7.6.2 Representation of the Individuals — stimulus map

This is the important result of the function. It represents the stimuli in the MCA factor space.

Element	Meaning
Two close stimuli	They were often placed in the same groups
Two distant stimuli	They were rarely grouped together
Compact group of stimuli	Family of stimuli often sorted together

The percentage displayed on each axis indicates the share of variance explained. Without verbalizations, this map remains more difficult to name: in that case, rely more on the category map and the automatic description of the dimensions.

7.6.3 Representation of the Categories — category map

This map represents the categories derived from the sorting variables. Categories close to stimuli are more strongly associated with them. The **Active categories**, **Supplementary categories**, and **Selection of categories** options control what is displayed.

> **Point of vigilance:** with many subjects, this map can become very crowded. The default value **cos2 10** helps keep only a more readable subset of categories.

7.6.4 Representation of the Variables — variable map

This map represents the qualitative variables of the MCA, that is, the subjects or sorting variables. It helps show to what extent the partitions of different subjects are aligned with the consensus axes. A subject with high coordinates on an axis has strongly structured their partition according to that axis.

7.6.5 Eigenvalue Decomposition — eigenvalue table

Element	Meaning
Dim.	Dimension number
Eigenvalue	Eigenvalue
% of variance	Share explained by the dimension
Cumulative %	Cumulative share explained

If the first two dimensions explain a large share of the variance, the 1/2 plane summarizes the structure of the sorting data well. Otherwise, explore other factor planes.

7.6.6 Automatic Description of the Dimensions

This textual output identifies the categories and variables most associated with each dimension, according to **Significance threshold (%)** and **Number of dimensions**. It is essential for giving sensory meaning to the axes, especially in the absence of verbalizations.

7.6.7 Representation of the Individuals According to Clusters *(available if "Representation of the clusters" is activated)*

This graph represents the stimuli on the factor map, grouped according to the clusters derived from the HCA.

Element	Meaning
Stimuli in the same cluster	Stimuli often perceived as close
Clearly separated clusters	More differentiated families of stimuli

7.6.8 Numerical tables *(optional)*

The coordinate, contribution, and cosine tables for stimuli and categories appear only if you check the corresponding options in **Numerical Indicators**. They provide precise values to complement the visual interpretation.

7.7 Example: the perfumes_sorting dataset

The **perfumes_sorting** dataset concerns 12 perfumes grouped by 60 subjects during a free sorting task. It is used to illustrate the function outputs. Some groups were also verbalized, which allows the function to produce both a factorial representation of the perfumes and a textual description of the profiles. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

7.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimuli Labels	variable identifying the 12 perfumes
Subjects	the first 50 sorting columns
Supplementary Subjects	from column 51 to 58
Ventilation level (%)	5
X-axis / Y-axis	1 / 2
Significance threshold (%)	5
Number of dimensions	2
Level description (%)	5
Number of components	5
Representation of the clusters	checked to visualize the classification
Number of clusters	-1

7.7.2 Analysis of the *Description of the Stimuli* table

The table shows that several perfumes have very marked lexical profiles. A few particularly clear readings, with the actual values from the analysis:

Aromatics Elixir is strongly associated with *strong* (test value 4.08, $p < .001$; 14.16% of internal occurrences versus 4.65% overall), *ammonia* (test value 2.26), and *spicy* (2.18). Conversely, it is clearly under-represented by *fruit* (test value -2.63), *flower* (-2.40), and *sweet* (-2.10): it is the most intense and least floral or sweet perfume in the panel.

Chanel N5 stands out with *soap* (test value 3.72, $p < .001$; 7.69% versus 1.61% overall), *deodorant* (2.06), and *unpleasant* (1.99), while being under-represented by *fruit* (-2.47) and *flower* (-2.21).

Lolita Lempicka is characterized by *sweet* (test value 3.94, $p < .001$; 14.55% versus 4.98% overall) and *vanilla* (2.09), but is under-represented by *flower* (-2.34).

Pleasures is the perfume most associated with *flower* (test value 3.46, $p < .001$; 15.00% versus 5.70% overall) and *fresh* (2.05), and clearly opposed to *strong* (-2.46).

Shalimar is characterized by *old* (test value 2.67, $p = 0.008$) and *sun* (2.40), and under-represented by *flower* (-2.42).

Angel is associated with *vanilla* (2.23), *heady* (1.99), and *too* (1.99), but under-represented by *fresh* (-2.03) and *flower* (-2.19).

7.7.3 Analysis of the *Representation of the Individuals* graph

The map reveals a readable perceptual structure on the Dim. 1 (15.53%) / Dim. 2 (14.14%) plane. On the right side of the plane, *Shalimar*, *Aromatics Elixir*, and *Chanel N5* form a group — confirming their strong, spicy, and low-floral profile. On the lower left, *J'adore ET*, *J'adore EP*, and *Pleasures* group together around a floral and fresh profile. At the top, *Angel* and *Lolita Lempicka* occupy a clearly distinct position on the second dimension — a vanilla and sweet profile. *Cinéma*, *L'instant*, and *Coco Mademoiselle* form an intermediate central-left area, with profiles that are more difficult to name without the textual description.

7.7.4 Analysis of the *Eigenvalue Decomposition* table

The first two dimensions explain 29.67% of the total variance (15.53% + 14.14%). This is clearly less concentrated than in a classical PCA: in free sorting, subjects have very different ways of structuring the space, which spreads the information across many dimensions. The 1/2 plane is a useful but partial summary — as confirmed by the following dimensions (Dim. 3: 11.03%, Dim. 4: 9.97%, etc.).

7.7.5 Analysis of the *Automatic Description of the Dimensions* table

On **Dim. 1**, the categories most strongly associated with the positive side (right, strong perfumes) are *solvent* (S52, estimate 1.31), *unpleasant* (S46, 1.27), *spicy* (S40, 1.24), *very;strong;perfume;woman;old* (S15, 1.32), *soap;ancient* (S48, 1.26), and *acid* (S54, 1.22). The negative side is associated with *flower;fresh* (S43, -0.60), *fresh;light* (S23, -0.81), and *flower* (S47, -0.68). Dim. 1 therefore clearly opposes **strong, spicy, acidic, or unpleasant** perfumes to **fresh and floral** perfumes.

On **Dim. 2**, the positive side is dominated by *vanilla;spicy;hot* (S16, 1.51), *woman;mature* (S7, 1.55), *sweet;too;strong* (S11, 1.43), *teenage;likeable* (S51, 1.54), and *vanilla;sweet* (S34, 1.05). The negative side includes *soap* (S12, -1.03) and *fresh* (S10, -0.50). Dim. 2 therefore opposes **vanilla, sweet, enveloping** perfumes to **fresher or soapy** profiles.

7.7.6 Analysis of the *Representation of the Individuals According to Clusters* graph

With **Number of clusters = -1**, the automatic classification retains **six clusters**:

- a cluster grouping **J'adore ET**, **J'adore EP**, and **Pleasures** (floral, fresh);
- a cluster grouping **Coco Mademoiselle**, **L'instant**, and **Cinéma** (intermediate profiles);
- a cluster containing only **Pure Poison** (singular profile);

- a cluster grouping **Angel** and **Lolita Lempicka** (vanilla, sweet);
- a cluster containing only **Chanel N5** (soapy, unpleasant);
- a cluster grouping **Shalimar** and **Aromatics Elixir** (strong, spicy, old).

7.7.7 Synthetic interpretation

This example clearly shows the value of the function. It does not only provide a proximity map between perfumes: it also shows **which words** subjects associate with each perfume, **which oppositions structure the space**, and **which groups of perfumes** emerge from the classification. Cross-reading the description table, the factor map, and the classification makes it possible to name the families more precisely than with the map alone.

> **What we take away from this example:** free sorting analysis makes it possible to link the factorial structure of the categorizations, their verbal interpretation, and the formation of coherent families of stimuli.

7.8 Practical tips

- Always check that all columns in **Subjects** are categorical before running the analysis.
- Use **Stimuli Labels** almost systematically to obtain readable maps.
- Start with **Ventilation level (%) = 5**.
- If the category map is too crowded, keep **Selection of categories = cos2 10** and uncheck **Active categories** if necessary.
- Leave **Number of clusters = -1** for automatic determination at first.
- Keep **Number of components = 5** for a first clustering.
- If the first two dimensions explain little variance, explore other factor planes.
- Activate **Coordinates** and **Cluster variable** if you want to reuse the results in other jamovi analyses.

7.9 In summary

This analysis transforms free sorting data into a reading of the stimuli that is simultaneously **factorial**, **textual**, and **typological**. It combines an **MCA** on the categorizations provided by the subjects, a **description of the stimuli based on the labels**, and, if desired, an **automatic classification** of the stimuli. Its main value is to preserve subjects' individual information while constructing a consensus and directly interpretable view of the perceptual space of the stimuli.

8 Hedonic Data — Preference Mapping

8.1 Introduction

This analysis makes it possible to construct a **preference map** from two types of information: an already computed **two-dimensional representation space**, and several **Liking Variables** corresponding to liking scores. It is used to identify which areas of this space are associated with stronger preference, and to visualize **what proportion of subjects** would theoretically be attracted to each region of the plane. In SEDA, this principle can be used both for **external preference mapping** (sensory space) and for **internal preference mapping** (hedonic space), depending on the origin of the two axes used.

Why not use a standard alternative?

A mean liking score by product makes it possible to compare products with one another, but it does not make it possible to relate preferences to an already constructed representation space. Here, the function fits a preference model **for each liking variable**, then combines these models into a **surface** indicating the **% of consumers** associated with each area of the plane. This provides a spatial reading of preferences, which is more informative than a simple table of means.

8.2 Understanding what happens under the hood

8.2.1 The function does not construct the representation space itself

The function does not calculate the axes of the plane: it expects you to provide one variable for **X-axis** and one variable for **Y-axis**. The integrated help text explicitly specifies the recommended workflow: for an external map, these axes may come from a PCA on sensory attributes; for an internal map, they may come from a PCA on hedonic scores. In both cases, the two components must first have been **saved** in the data, then selected in this module.

How to retrieve dimensions from the MEDA module

1. Open the **MEDA** module (menu **MEDA > PCA**).
2. Place your variables in the appropriate fields: for an external map, use your sensory attributes; for an internal map, use your hedonic variables.

3. Expand the **Numerical Indicators** section, then the **Save** subsection.
4. In the **Number of components** field, enter **2** (or more if you want to explore other planes).
5. Activate the **Coordinates** output: jamovi then automatically creates two new columns in your data sheet, named by default **Dim. 1** and **Dim. 2**, containing the coordinates of each stimulus on the first two components.
6. Then return to **SEDA > Hedonic Data > Preference Mapping** and place **Dim. 1** in **X-axis** and **Dim. 2** in **Y-axis**.

If your dataset contains the sensory attributes, the hedonic scores, and the stimulus identifiers in the same file — as is the case for the **senso_hedo_cocktail** dataset — this workflow can be performed entirely in jamovi without any intermediate export or import.

8.2.2 A model is fitted for each liking variable

The function first gathers the two coordinates of the plane and all the columns placed in **Liking Variables**. It then fits a separate model for each liking variable. The type of model depends on **Regression Model**:

- **Quadratic** (default): linear effects, quadratic terms, and interaction between the two axes. This is the most flexible model. It requires a sufficient number of products — at least 7, ideally 10 or more — to be fitted reliably.
- **Vector**: linear effects only. It looks for a global direction of preference in the space.
- **Circular**: linear effects with a common radial quadratic structure. It looks for a more centered optimum.
- **Elliptical**: separate quadratic effects on each axis, without interaction. It allows an optimal area elongated in certain directions.

A simple way to understand them is as follows: **Vector** assumes a monotonic trend; **Quadratic** allows a flexible optimal area; **Circular** and **Elliptical** assume a more or less concentrated local optimum.

8.2.3 Construction of the preference surface

Once the models have been fitted, the function constructs a **regular grid** on the plane defined by **X-axis** and **Y-axis**. For each liking variable, it calculates the predicted values on this grid, centers these predictions, then counts at each point how many subjects exceed a certain reference level. The final result is converted into a **percentage** and displayed on the map as **% of consumers**.

8.2.4 What the final map really represents

The final map does not directly show the observed scores product by product. It shows a **modeled surface**, constructed from the individual models fitted on the plane. The colored areas indicate regions of the space where a larger proportion of liking variables is predicted above the reference level. This explains the legend label **% of consumers**.

8.3 Expected data structure

The data must contain:

- one optional column for **Stimuli Labels**;
- one numeric column for **X-axis**;
- one numeric column for **Y-axis**;
- at least **two** numeric columns in **Liking Variables**.

Each row corresponds to a **stimulus**. The two coordinates represent its position in the 2D space, and the columns in **Liking Variables** contain the scores associated with that same stimulus. The function requires at least **2** liking variables; otherwise, it rejects the analysis.

Before running the analysis in jamovi, check in the **Data** tab that:

- **Stimuli Labels** is categorical if you use it;
- **X-axis** and **Y-axis** are numeric;
- all columns in **Liking Variables** are numeric;
- you have selected at least **two** variables in **Liking Variables**.

8.4 Interface guide

The analysis appears in **SEDA > Hedonic Data > Preference Mapping**. The interface includes four variable drop zones, the **Read me before running** checkbox, the **Regression Model** menu, and then two color menus in **Graphic Options**.

8.4.1 Variables to complete

8.4.1.1 Stimuli Labels

Role: provides the names displayed for the stimuli on the map. **Impact on the computation:** it does not change the calculation of the surface, but determines the labels plotted next to the points. The function automatically replaces missing labels with numeric identifiers.

Practical tip: use short and distinct labels; otherwise, the map quickly becomes difficult to read.

8.4.1.2 X-axis

Role: provides the horizontal coordinate of the stimuli in the representation space. **Impact on the computation:** it directly enters the regression models and defines the first axis of the surface. **Practical tip:** use an already saved dimension, typically the first component from a PCA previously performed in the MEDA module.

8.4.1.3 Y-axis

Role: provides the vertical coordinate of the stimuli in the representation space. **Impact on the computation:** it enters the regression models and defines the second axis of the surface. **Practical tip:** use the second component from the same space as the one chosen for **X-axis**.

8.4.1.4 Liking Variables

Role: groups together the liking scores used to construct the map. **Impact on the computation:** the function fits a separate model for each variable, then aggregates the results into the final surface. The more variables you have, the more faithfully the surface represents a percentage of subjects. **Practical tip:** use the available individual scores, not only means. At least **2 variables** are required.

8.4.2 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

8.4.3 Regression Model

Role: chooses the form of the model fitted for each liking variable. **Impact on the computation:** this option directly modifies the shape of the preference surface.

Model	Description	Recommended use
Quadratic (default)	Linear and quadratic effects, plus interaction	General case; ideally 10 products

Model	Description	Recommended use
Vector	Linear effects only	Few products or monotonic preference
Circular	Common radial quadratic structure	Assumed centered optimum
Elliptical	Separate quadratic effects by axis, without interaction	Assumed elongated optimum

Practical tip: start with **Quadratic** (default value). If you have fewer than 10 products, prefer **Vector** or **Circular**.

8.4.4 Graphic Options

8.4.4.1 Color for low preference areas

Role: defines the color of low-preference areas. **Impact on the computation:** no effect on the models; it only changes the graphical palette. **Practical tip:** the default value is **Blue**. The palette used is a gradient from this color to white, then to the high-preference color.

8.4.4.2 Color for high preference areas

Role: defines the color of high-preference areas. **Impact on the computation:** no effect on the models; it only changes the palette. **Practical tip:** the default value is **Red**. Keeping **Blue** for low areas and **Red** for high areas is generally the most readable combination.

8.5 Step-by-step procedure

1. Open your file in jamovi.
2. If you do not yet have the coordinates of the 2D space, first construct them in the **MEDA** module: run a **PCA**, activate **Coordinates** in the **Save** subsection of **Numerical Indicators**, and enter **2** in **Number of components**. Two new columns (**Dim. 1** and **Dim. 2**) automatically appear in your data sheet.
3. In the **Data** tab, check that the two axes and the liking variables are numeric.
4. Open **SEDA > Hedonic Data > Preference Mapping**.
5. Place the stimulus name variable in **Stimuli Labels**.
6. Place the first saved component in **X-axis**.
7. Place the second saved component in **Y-axis**.
8. Place all score columns in **Liking Variables** (at least **2**).

9. Leave **Regression Model = Quadratic** for a first trial.
10. Leave the default colors (**Blue / Red**) if you do not want to customize the palette.
11. Read the map by identifying the most favorable areas and the positions of the stimuli.
12. Optionally test another **Regression Model** to see whether the interpretation remains stable.

8.6 Interpretation of results

8.6.1 Preference Mapping — main graph

The function produces a single output: the preference map. It represents the provided factor space as a background, over which a modeled density surface is superimposed.

Element	Meaning
Areas toward **Color for high preference areas** (default: red)	Regions where a high percentage of subjects would have an estimated favorable liking
Areas toward **Color for low preference areas** (default: blue)	Regions with low liking
White areas	Intermediate regions
Contour lines	Indicate the exact percentage of consumers (0 to 100)
Tight contour lines	Rapid change in the percentage in this area
Squares with labels	Position of the tested stimuli in the factor space

The area closest to the high-preference color corresponds to the region of the plane where the estimated proportion of "favorable" subjects is highest. It can be interpreted as an **optimal preference area** in the chosen space.

> **Point of vigilance:** the map is based on modeling. It does not only show the observed products, but an interpolation over the whole plane. Be cautious when interpreting areas far from the stimuli actually present: extrapolation is less reliable there.

> **Point of vigilance:** the surface depends both on the number of stimuli available to fit the models and on the number of liking variables used to construct the percentage. A visually smooth map does not in itself guarantee high robustness.

8.6.2 Absence of numerical tables

In the current configuration of the module, no numerical results table is produced. Interpretation is therefore based entirely on the **Preference Mapping** graph.

8.7 Example: the senso_hedo_cocktail dataset

The **senso_hedo_cocktail** dataset contains the sensory evaluations of 13 cocktails described by 13 attributes, as well as hedonic judgments from a panel of 47 subjects. It is used to illustrate the construction of a **preference map**: the sensory space is first constructed by PCA in MEDA, then the hedonic scores are projected into it through this function. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

8.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimuli Labels	variable identifying the cocktails
X-axis	Dim. 1 (saved from the PCA)
Y-axis	Dim. 2 (saved from the PCA)
Liking Variables	all hedonic columns (O to DJ)
Regression Model	Quadratic
Color for low preference areas	Blue
Color for high preference areas	Red

8.7.2 Step 1 — Construct the sensory space in MEDA (PCA)

In the **MEDA** module, run a PCA on the 13 sensory attributes with the following settings:

Option	Value
Active Variables	the 13 sensory attributes
Scale to unit variance	Checked
Number of components (Save)	2
Coordinates (Save)	Checked

The PCA produces a two-dimensional space: **Dim. 1 explains 53.25%** of the variance and **Dim. 2 explains 24.13%**, that is, **77.37% cumulative**. The 1/2 plane therefore summarizes the sensory profiles of the cocktails well.

The automatic description of the dimensions confirms the interpretation of the axes:

- **Dim. 1** opposes *acidity* (correlation 0.93), *bitterness* (0.88), *odor.orange* (0.75), and *intensity* (0.73) on one side, to *odor.banana* (-0.93), *sweet* (-0.93), *thickness* (-0.88), and *pulp* (-0.70) on the other. This is a clear opposition between acidic and intense cocktails vs sweet, thick, banana-like cocktails.
- **Dim. 2** is more strongly structured by *odor.lemon* (0.77), *odor.mango* (0.69), *persistence* (0.66), and *color.intensity* (0.58), with *odor.orange* (-0.61) on the negative side.

On the individual map, the cocktails are widely distributed across the plane. **386** is located in the upper right (high on Dim. 2), **131** at the bottom center, **447** and **312** on the left (sweet, thick), and **194**, **233**, and **245** on the right (acidic, intense). Once the PCA has been run, the **Dim. 1** and **Dim. 2** columns automatically appear in the jamovi data sheet.

8.7.3 Step 2 — Construct the preference map in SEDA

Return to **SEDA > Hedonic Data > Preference Mapping** and use the following settings:

Option	Value
Stimuli Labels	variable identifying the cocktails
X-axis	Dim. 1 (saved from the PCA)
Y-axis	Dim. 2 (saved from the PCA)
Liking Variables	all hedonic score columns (O to DJ)
Regression Model	Quadratic
Color for low preference areas	Blue
Color for high preference areas	Red

8.7.4 Analysis of the *Representation of the products* graph

The resulting map superimposes the density surface onto the position of the 13 cocktails in the sensory space. The dark red area — corresponding to more than 80–90% favorable consumers — is located **on the left of the plane**, in the region with negative coordinates on Dim. 1, where the cocktails **182**, **444**, **343**, and **447** are located. These cocktails, characterized by sweeter, thicker, banana-like profiles, correspond to the optimal preference area.

Conversely, the blue area — corresponding to fewer than 20% favorable consumers — is located **on the right of the plane**, where the cocktails **194**, **233**, and **245** are located, characterized by acidic and intense profiles.

The cocktails **131** (bottom center) and **386** (upper right) are located in intermediate to low-preference areas (light blue, 10–25%).

8.7.5 Synthetic interpretation

The sensory space structured by the PCA, based on the acidic-intense vs sweet-thick opposition, is directly readable on the preference map. The optimum is located on the sweet-thick side, whereas acidic and intense profiles are less appreciated. No cocktail is located exactly in the maximum preference area, which suggests more of a **formulation improvement direction** than an already optimal product.

> **What we take away from this example:** preference mapping directly links the sensory space and hedonic preferences, making it possible to identify an optimal area even when no existing product occupies it exactly.

8.8 Practical tips

- Always perform a preliminary PCA in the MEDA module and save the components before using this module.
- Check that **X-axis** and **Y-axis** come from the same representation space.
- Use individual scores in **Liking Variables**, not only means.
- At least **two** variables are required in **Liking Variables** to produce the map.
- Start with **Regression Model = Quadratic**; if you have fewer than 10 products, test **Vector** or **Circular**.
- Do not interpret too far from the products actually present on the plane.
- An optimal area outside the existing products mainly indicates a direction for formulation or development.
- Keep a contrasted palette to clearly distinguish low and high areas.

8.9 In summary

This analysis constructs a **preference map** from an already available 2D space and several hedonic variables. It fits a preference model for each variable in **Liking Variables**, then combines these models into a surface expressed as **% of consumers**. Its main value is to link, within the same framework, a representation space and individual preferences, in order to identify the areas of the plane that are most favorable for the greatest number — and thus guide product development toward the most appreciated characteristics.